

SRPS to eSRPS Upgrade



OPERATING DOCUMENTATION

5670089
Revision 1.0



WARNING



STRONG MAGNETIC FIELD



NO PACEMAKERS*
NO NEUROSTIMULATORS*
NO CONDUCTIVE/METALLIC IMPLANTS*



**Persons with pacemakers,
neurostimulators or metallic
implants must not enter this area.**

Serious injury may result.

* In general, patients with conductive (e.g. metallic) implants are contraindicated for MR scans. For patients with implants that are labeled as 'MR Safe' or 'MR Conditional', consult the implant device manufacturer's documentation.

* WARNING: Only use quadrature transmit for 'MR Conditional' devices.

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Important Information

LANGUAGE

ПРЕДУПРЕЖДЕНИЕ (BG)

Това упътване за работа е налично само на английски език.

- Ако доставчикът на услугата на клиента изиска друг език, задължение на клиента е да осигури превод.
- Не използвайте оборудването, преди да сте се консултирали и разбрали упътването за работа.
- Неспазването на това предупреждение може да доведе до нараняване на доставчика на услугата, оператора или пациента в резултат на токов удар, механична или друга опасност.

警告 (ZH-CN)

本维修手册仅提供英文版本。

- 如果客户的维修服务人员需要非英文版本，则客户需自行提供翻译服务。
- 未详细阅读和完全理解本维修手册之前，不得进行维修。
- 忽略本警告可能对维修服务人员、操作人员或患者造成电击、机械伤害或其他形式的伤害。

警告 (ZH-HK)

本服務手冊僅提供英文版本。

- 倘若客戶的服務供應商需要英文以外之服務手冊，客戶有責任提供翻譯服務。
- 除非已參閱本服務手冊及明白其內容，否則切勿嘗試維修設備。
- 不遵從本警告或會令服務供應商、網絡供應商或病人受到觸電、機械性或其他危險。

警告 (ZH-TW)

本維修手冊僅有英文版。

- 若客戶的維修廠商需要英文版以外的語言，應由客戶自行提供翻譯服務。
- 請勿試圖維修本設備，除非 您已查閱並瞭解本維修手冊。
- 若未留意本警告，可能導致維修廠商、操作員或病患因觸電、機械或其他危險而受傷。

UPOZORENJE (HR)

Ovaj servisni priručnik dostupan je na engleskom jeziku.

- Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod.
- Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik.
- Zanimarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.

**VÝSTRAHA
(CS)**

Tento provozní návod existuje pouze v anglickém jazyce.

- V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka.
- Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah.
- V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.

**ADVARSEL
(DA)**

Denne servicemanual findes kun på engelsk.

- Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse.
- Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual.
- Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for teknikeren, operatøren eller patienten.

**WAARSCHUWING
(NL)**

Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar.

- Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan.
- Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is.
- Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

**WARNING
(EN)**

This service manual is available in English only.

- If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services.
- Do not attempt to service the equipment unless this service manual has been consulted and is understood.
- Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.

**HOIATUS
(ET)**

See teenindusjuhend on saadaval ainult inglise keeles.

- Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest.
- Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist.
- Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.

**VAROITUS
(FI)**

Tämä huolto-ohje on saatavilla vain englanniksi.

- Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla.
- Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen.
- Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteen käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.

**ATTENTION
(FR)**

Ce manuel d'installation et de maintenance est disponible uniquement en anglais.

- Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire.
- Ne pas tenter d'intervenir sur les équipements tant que ce manuel d'installation et de maintenance n'a pas été consulté et compris.
- Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.

**WARNUNG
(DE)**

Diese Serviceanleitung existiert nur in englischer Sprache.

- Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen.
- Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben.
- Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendienst-technikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.

**ΠΡΟΕΙΔΟΠΟΙΗΣΗ
(EL)**

Το παρόν εγχειρίδιο σέρβις διατίθεται μόνο στα αγγλικά.

- Εάν ο τεχνικός σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει τις υπηρεσίες μετάφρασης.
- Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό αν δεν έχετε συμβουλευτεί και κατανοήσει το παρόν εγχειρίδιο σέρβις.
- Αν δεν προσέξετε την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στον τεχνικό σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.

**FIGYELMEZTETÉS
(HU)**

Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el.

- Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése.
- Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték.
- Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.

AÐVÖRUN
(IS)

Þessi þjónustuhandbók er aðeins fáanleg á ensku.

- Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálaþjónustu.
- Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin.
- Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.

AVVERTENZA
(IT)

Il presente manuale di manutenzione è disponibile soltanto in lingua inglese.

- Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.
- Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto.
- Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.

警告
(JA)

このサービスマニュアルには英語版しかありません。

- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

경고
(KO)

본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다.

- 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다.
- 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오.
- 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.

BRĪDINĀJUMS
(LV)

Šī apkopes rokasgrāmata ir pieejama tikai angļu valodā.

- Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu.
- Neveiciet aprikojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas.
- Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumu risks apkopes sniedzējam, operatoram vai pacientam.

**[SPĖJIMAS
(LT)**

Šis eksploatavimo vadovas yra tik anglų kalba.

- Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas.
- Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo.
- Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.

**ADVARSEL
(NO)**

Denne servicehåndboken finnes bare på engelsk.

- Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse.
- Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått.
- Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.

**OSTRZEŻENIE
(PL)**

Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim.

- Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta.
- Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go.
- Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.

**ATENÇÃO
(PT-BR)**

Este manual de assistência técnica encontra-se disponível unicamente em inglês.

- Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.

**ATENÇÃO
(PT-PT)**

Este manual de assistência técnica só se encontra disponível em inglês.

- Se qualquer outro serviço de assistência técnica solicitar este manual noutra idioma, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.

**ATENȚIE
(RO)**

Acest manual de service este disponibil doar în limba engleză.

- Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere.
- Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service.
- Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.

**ОСТОРОЖНО!
(RU)**

Данное руководство по техническому обслуживанию представлено только на английском языке.

- Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод.
- Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения.
- Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.

**UPOZORENJE
(SR)**

Ovo servisno uputstvo je dostupno samo na engleskom jeziku.

- Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge.
- Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo.
- Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.

**UPOZORNENIE
(SK)**

Tento návod na obsluhu je k dispozícii len v angličtine.

- Ak zákazník poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka.
- Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu.
- Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.

**ATENCION
(ES)**

Este manual de servicio sólo existe en inglés.

- Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual.
- No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio.
- La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.

**VARNING
(SV)**

Den här servicehandboken finns bara tillgänglig på engelska.

- Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster.
- Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken.
- Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

**OPOZORILO
(SL)**

Ta servisni priročnik je na voljo samo v angleškem jeziku.

- Če ponudnik storitve stranke potrebuje priročnik v drugem jeziku, mora stranka zagotoviti prevod.
- Ne poskušajte servisirati opreme, če tega priročnika niste v celoti prebrali in razumeli.
- Če tega opozorila ne upoštevate, se lahko zaradi električnega udara, mehanskih ali drugih nevarnosti poškoduje ponudnik storitev, operater ali bolnik.

**DİKKAT
(TR)**

Bu servis kılavuzunun sadece ingilizcesi mevcuttur.

- Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

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Revision History

Revision	Date	Description
1.0	May 28, 2014	Initial release.

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Chapter 1 Overview

1 Getting Started

1.1 Upgrade Introduction

This publication documents the mechanical installation information for upgrading a system with SRPS to eSRPS.



WARNING

FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES.

KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

1.2 Site Ready Check

Pre-installation work must be completed before equipment is delivered to avoid delays and confusion.

1.3 Pre-Delivery FDO Checks

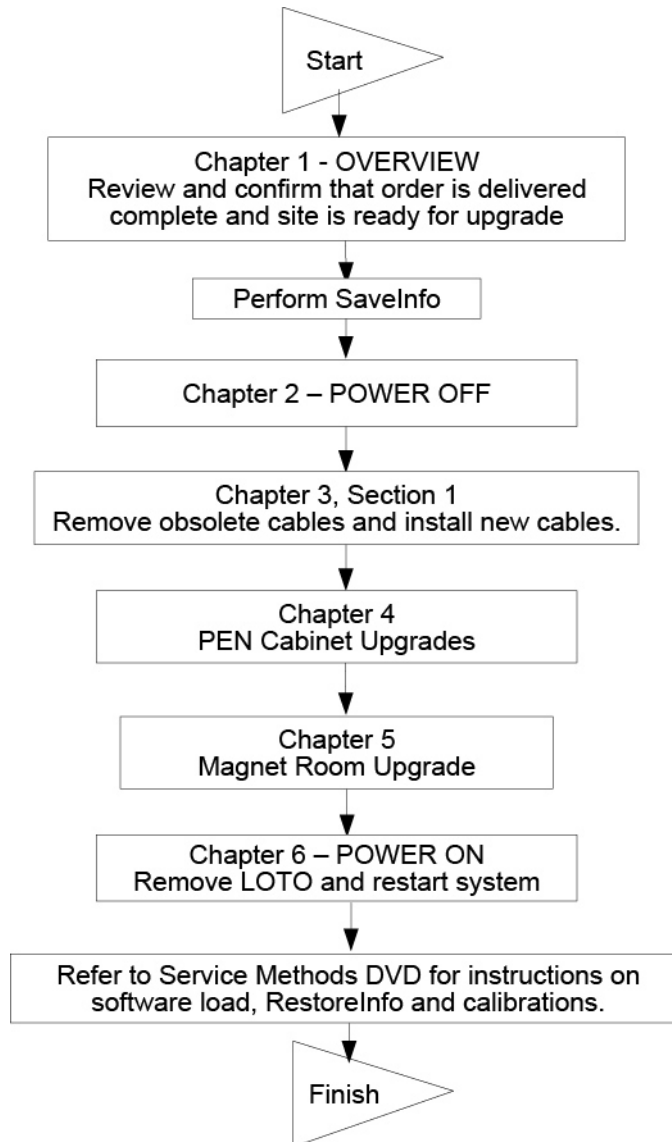
Before the shipment is made, Be sure that FDO review has been completed by consultation with the Upgrade Specialist. Also verify the Surface coil compatibility issues have also been resolved.

1.4 Upgrade Schedule Considerations

FMs, periodic maintenance, installation of prerequisite upgrades and/or options will directly add to the man-hour and system down-time total. Be sure that all planned activity is considered when predicting the return of system to customer.

1.5 Upgrade Flowchart

Illustration 1-1: eSRPS Upgrade Flowchart



1.6 Product Delivery Instructions and Packing

The "Shipping Document" lists catalog numbers delivered. **Review and confirm that order is delivered complete.** Check impact on installation schedule if Catalogs and/or packing boxes are missing and/or noted as shipped short.

Labels that are attached to the outside of packing boxes summarize box contents. PDI (Product Delivery Instructions) specify box contents, part numbers, and shipping procedure. The PDI is numbered according to the catalog number. Lists of items included with each box are detailed by separate checklists, or a separate sheet that provides a summary of that box contents. Refer to PDI and packing lists for information specific to your shipment.

A set of service and operator manuals is delivered with the Upgrade Kit. Refer to checklists packed with "Technical Publication" boxes for a list of delivered documents.

1.7 Damage in Transportation

All packages should be closely examined at time of delivery. If damage is apparent, have notation "**damage in shipment**" written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period. File a report with:

- Call 1-800-548-3366. Select "Install Support Services for FOA and MIS"
- Contact your local service coordinator for more information on this process.

1.8 Product Locator

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans-shipment, and field location of the serialized models. There are now two methods for submitting Product Locator information.

1. At this time, for **U.S. ONLY**, the preferred method of submitting information is the FE Site Verification Web Site at:

<http://supportcentral.ge.com/15563>

The FE Site Verification consists of three components that are available on the web from the main menu. They are:

- Install/deinstall product locator model and serial numbers
- Add/modify ship to address information
- Update CARES FE data for primary/secondary FEs

To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following: Product Locator Support Central Page - <http://supportcentral.ge.com/15563>

2. One "Shipping Card" is filled out and submitted when shipped (extra cards are supplied for trans-shipments between storage and distribution points), and the "Installation Card" and extra shipment cards are attached.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

1.9 Required Equipment and Tools



NOTICE

Follow all required safety and PPE procedures customary for your organization when working on this product.

Table 1-1: General Safety Requirements

Item	Description
1	Wear composite safety shoes for foot protection.
2	Wear gloves for hand protection.
3	Wear safety glasses for eye protection.



DANGER

FERROUS MATERIAL HAZARD!

IF MAGNET IS ENERGIZED, TOOLS, COMPONENTS, AND PARTS REQUIRED FOR THIS INSTALLATION THAT CONTAIN FERROUS MATERIAL WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES.

KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

Tools to upgrade the system are listed in following table:

Table 1-2: Installation Equipment

Item	GE Part Number	Description
1	5112581	Non-Magnetic Service Tool Kit
2	46-271138G1	Restricted Access Control Kit. Contains two plastic warning signs for posting at site during installation and service activity.
3		Service Methods DVD
4	46-198094P1	Wrist grounding strap
5	---	Plastic or aluminum flashlight
6	5182417	MCRv Tool Kit
7		Assorted Cable Ties

1.10 Discarded Material Return Policy

The equipment or material being removed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

1.10.1 Material Policy Overview

1.10.1.1 Returned Material

Per GE Healthcare Policy 3.9: all requested material is to be returned, as directed, intact and promptly to Renewable Resources. Do not remove or swap any parts from the material to be returned.

1.10.1.2 Disposed Material

Regulations governing the disposal of this equipment or material vary from location to location and from country to country. In order to insure compliance with all local, state, or national environment, health, and safety regulations, dispose of all items per local GEMS policy and procedures and other applicable published guidance. Contact GEMS Recycling Center for assistance.

1.10.2 Material to be Returned

All deinstalled components must be returned to GE Renewable Resources for recycling.

- Mark each box: SRPS to eSRPS UPGRADE
- Send the boxes to the following address:

1.10.2.1 Sites in Americas

GE Renewable Resources
Attn: Asset Recovery Leader, Harvest
7624 South 10th Street
Oak Creek, WI 53154

1.10.2.2 Sites in Asia

Return procedure for GEMS–Asia countries, contact:

GE HEALTHCARE JAPAN CORPORATION
ATTN: SWaP/HARVEST
67-4, Takakura-machi, Hachioji-shi-Tokyo
192-0033 Japan
Contact at Destination: 81-426-48-2595
FAX: 81-426-48-2902

1.10.2.3 Sites in Europe

For non-Gold Seal returns in Europe:

European Renewable Resources Center
GEODIS LOGISTICS - CD 26 - Pièce de la Remise - Bâtiment EVL2
Quai 45 à 49 - 91090 LISSES - FRANCE
TEL = (+33)-1-64-97-55-41 FAX = (+33)-1-60-86-12-31

1.10.3 Return Kit

If needed, packaging and/or transportation details can be obtained through the local Installation Specialist.

Reuse appropriate containers that new hardware and modules were shipped to site for return of above material.

1.10.4 Deinstallation Procedures

Refer to instructions in this manual for deinstallation procedures of site.

1.10.5 Product Locator Information

Process product locator information for all serialized components removed from the system according to policy. Refer to [Section 1.8](#).

1.11 Upgrade Components

The following is an overview of components required for upgrading a system with SRPS to eSRPS.

- eSRPS Module
- Filter Box for eSRPS
- Pen Panel Access Panel
- ePDB Module
- RFSB2–3T Module
- A-Port Legacy Cable Track
- P1–P2 Cable Track
- ODU Dock Harness
- Upgrade Cables

Chapter 2 Power Off Procedures

1 Power-Off Procedure

Prior to system shut down, the LPCA should be moved to within 1 foot (30cm) of the end of the bridge on the rear pedestal.

Perform LOTO on the PDU cabinet. See Chapter 3, Section 2.4, of the *MR Service Safety Manual*, PN 5452735.

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Chapter 3 Cable Maps

1 Cable Interconnect Upgrade

NOTICE

This upgrade consists of the removal and installation of cables in the overhead tray. The MR Service Platform, part 5155291, must be used with the safety rails installed.

NOTICE

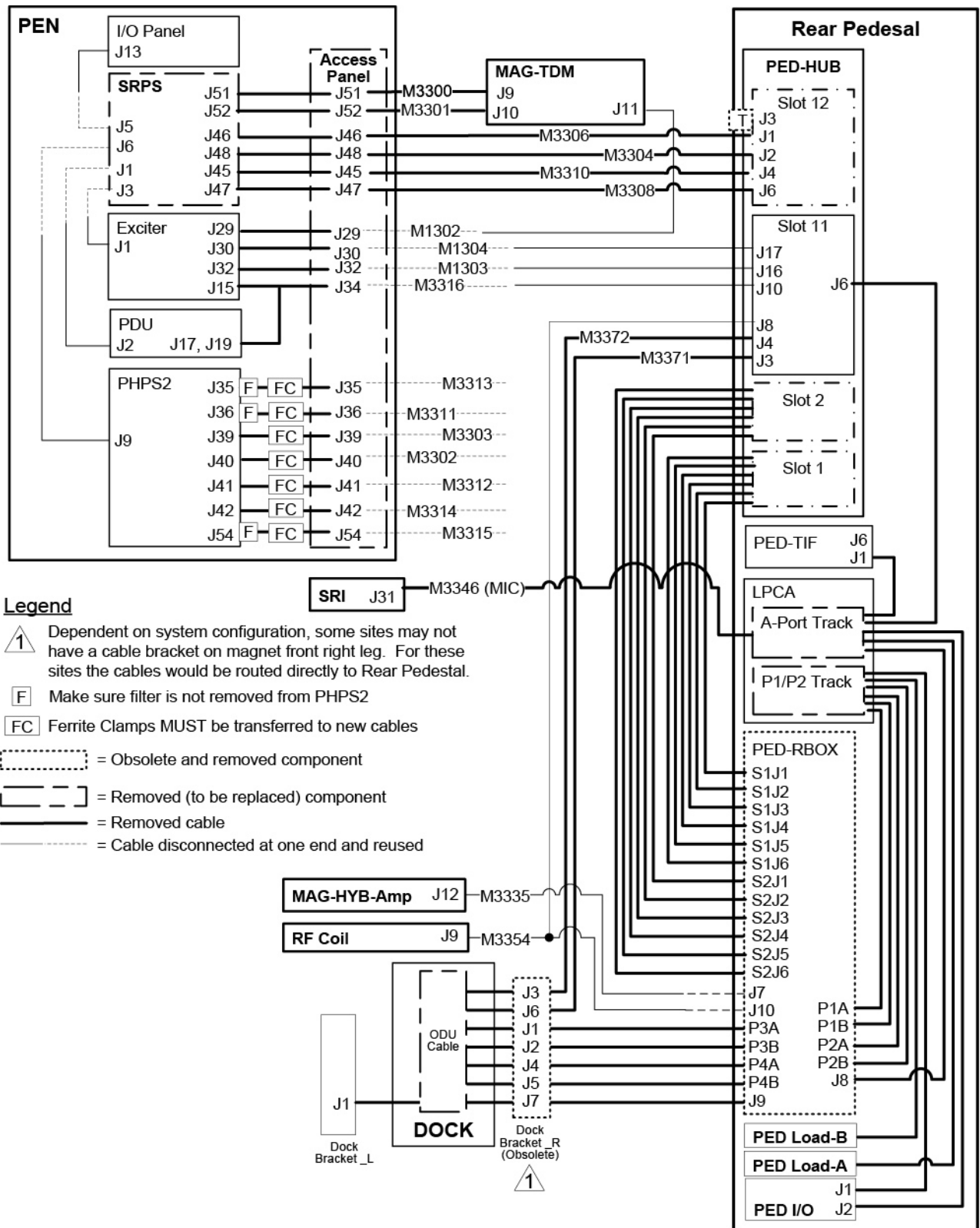
All obsolete RF Transmit cables **MUST** be removed when performing this upgrade to ensure good image quality, and successful Dual Drive Quadrature calibration. Failure to comply will result in calibration failure and unnecessary troubleshooting.

This upgrade will remove and replace specific cables and also add new cables. The details for the upgrade procedures are found in the sections that follow.

1.1 Cable and Component Removal

The following illustration not only shows the cables and components that will be removed, it also shows the cables that will be disconnected and reused.

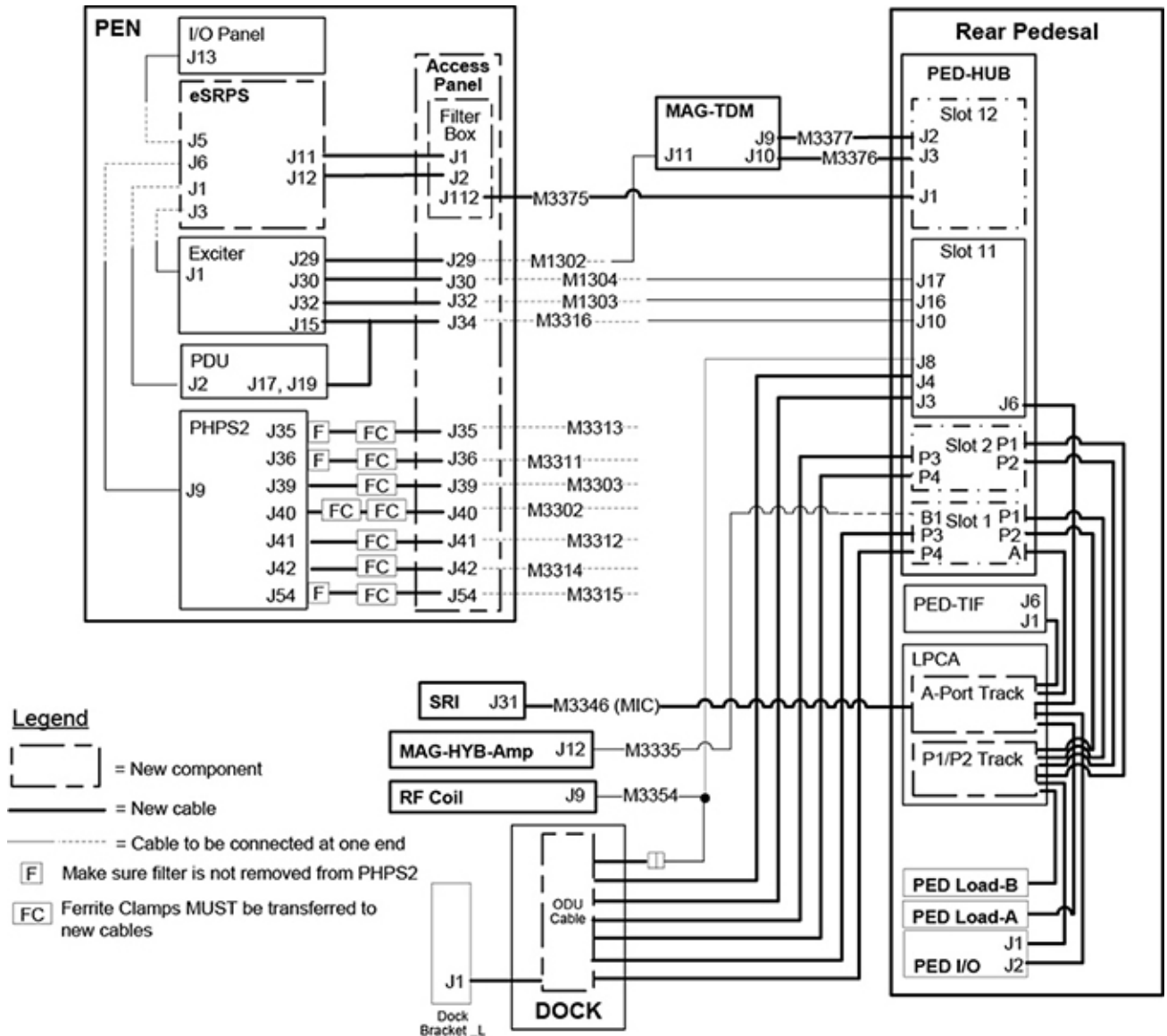
Illustration 3-1: Affected Cables and Components



1.2 New Cable and Component Installation

The illustration shows new cables to be routed and connected, existing cables to be reused, and new components to be installed.

Illustration 3-2: Cable and Component Installation



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Chapter 4 PEN Cabinet Upgrade Procedures

1 SRPS Module Removal

This chapter describes the procedure for replacing the Scan Room Power Supply (SRPS) with the EcoPower Scan Room Power Supply (eSRPS), the existing Access panel on the PEN Panel with the eSRPS Access Panel, and associated cables.

Because of its weight, removing the SRPS from the PEN cabinet requires you to use the hoist service tool. The hoist service tool allows one person to perform this procedure. This is the preferred method to avoid injury.

The eSPRS is light enough to be installed by one person without the use of a hoist.



WARNING

POSSIBLE PERSONAL INJURY OR EQUIPMENT DAMAGE!
THIS UNIT COULD FALL WHEN USING THE HOIST TOOL.
BEFORE PUTTING WEIGHT ON THE HOIST CHAIN, BE SURE THAT THE LINKS ARE ALIGNED PROPERLY. THIS HELPS KEEP THE CHAIN FROM BINDING UP, OR BECOMING KINKED, WHICH IS NEARLY IMPOSSIBLE TO CORRECT WHEN THE CHAIN IS UNDER TENSION.

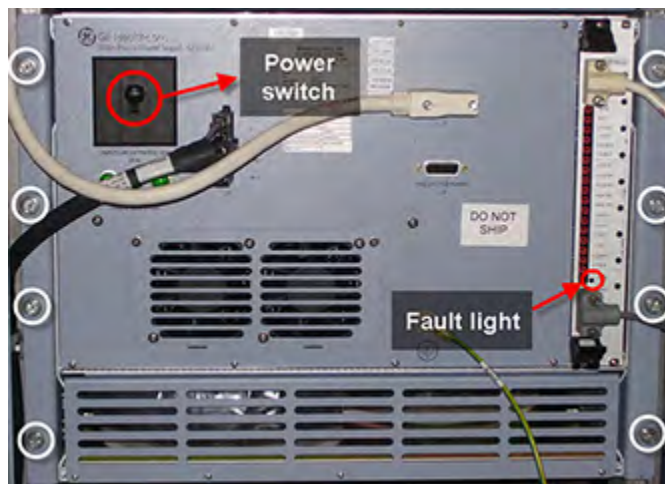


CAUTION

Make sure the proper Lockout/Tagout (LOTO) procedure for the PEN Cabinet has been performed. The cabinet should not have any power as you are assembling and using the hoist kit.

1. Disconnect the cables from the front of the SRPS.
2. Loosen the 8 thumb screws on the front of the SRPS.

Illustration 4-1: SRPS Front



3. In the HEC, remove the support tube. The support tube is located on the right hand side of the cabinet and is secured with two mounting bolts

There are seven openings in the support tube to be used in multiple configurations.

Illustration 4-2: Support Tube - Openings Labels

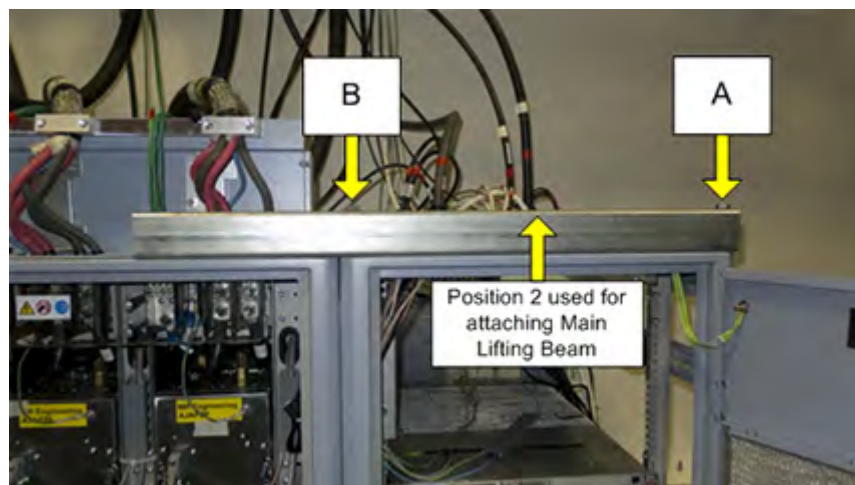


Table 4-1: Support Tube Orientation

Cabinet	Module	Left Cabinet Attachment	Main Lifting Beam Attachment	Right Cabinet Attachment
PEN (Support Tube Overhang Right)	SRPS	A	2	B

- Place the support tube on top of the PEN cabinet. To place the support tube correctly onto the cabinet, follow the proper orientation, see table below. Secure the support tube on top of the cabinet using a wrench to tighten the two bolts onto the top.

Illustration 4-3: Example: Support Tube in Position for RF Amplifier Removal



DANGER

ELECTROCUTION HAZARD!

THE LIFTING BEAM REQUIRED FOR THE NEXT STEP IS LOCATED IN THE PGR CABINET WHERE CABLES HAVE TO BE DISCONNECTED TO GAIN ACCESS TO THE BEAM.

MAKE SURE THAT LOTO PROCEDURES IN CHAPTER 3, SECTION 2.6 OF THE MR SERVICE SAFETY MANUAL, 5452735, HAVE BEEN PERFORMED BEFORE REMOVAL OF BEAM.

- Remove the main lifting beam from the PGR cabinet. The main lifting beam is located on the right side of the PGR cabinet and is secured with Velcro straps.

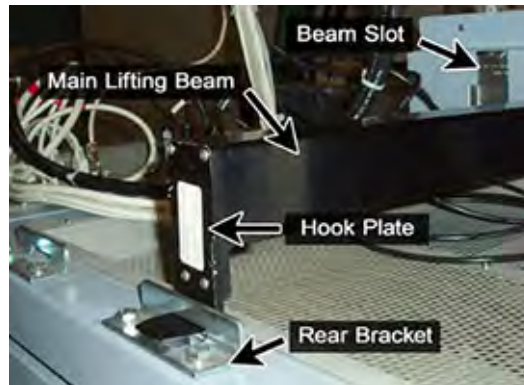
To take out the main lifting beam, you must unscrew and remove the connector from J9 on the Z-axis XPS unit.

- The main lifting beam slides onto the top of the cabinet. You may have to move cables or hoses out of the way, but none of the cables or hoses need to be disconnected, especially

for the PEN cabinet. There is a rear bracket for each position. The main lifting beam hook plate slides into the rear bracket. Check to ensure that the rear bracket is tight and secure.

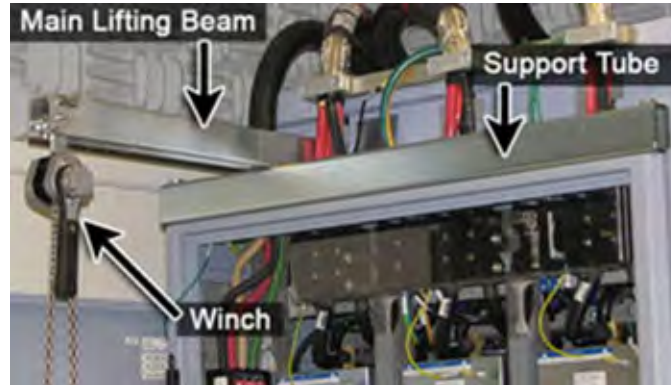
The bracket lines up with the support tube's orientation. If the bracket is not aligned properly to receive the main lifting beam, then the support tube orientation setup is incorrect for the cabinet.

Illustration 4-4: Cabinet Rear Bracket



7. Remove the winch from the Hoist Service Kit. Slide the winch into the main lifting beam. Secure the winch in with the pin so it does not slide out. The winch is able to slide within the main lifting beam to allow proper alignment with the removed item or FRU lifting bracket.

Illustration 4-5: Hoist Kit Assembled



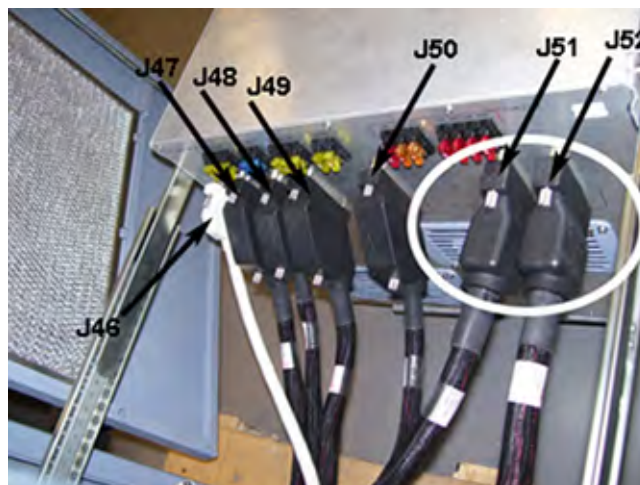
8. Pull the SRPS out from the PEN cabinet until the slide rails lock into position.

Illustration 4-6: Pull out SRPS



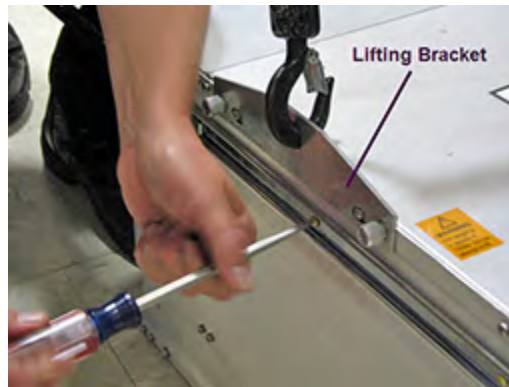
9. Disconnect the cables from the rear of the module.

Illustration 4-7: SRPS Rear with J51 and J52 Highlighted



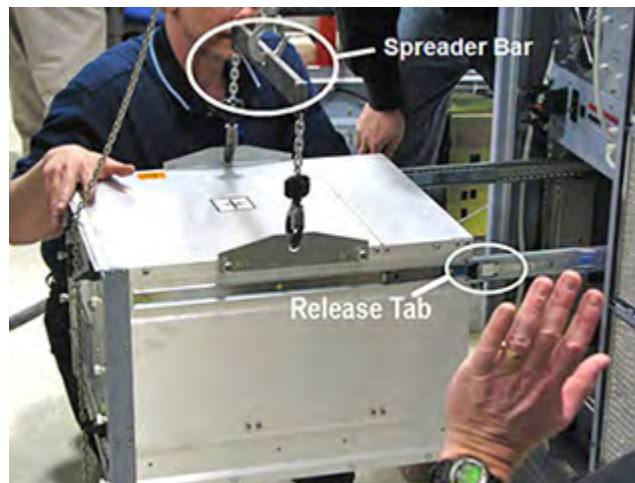
10. The other ends of cables J51 and J52 are attached to J51 and J52 on the bottom of the PEN Panel. These cables are to be disconnected and discarded.
11. Attach a lifting bracket from the FRU container to each side of the SRPS.

Illustration 4-8: Attach Lifting Brackets



12. Attach the spreader bar (hoist kit) to the lifting brackets. Attach the spreader bar to the hoist chain.

Illustration 4-9: Spreader Bar - Hoist Kit



13. Ratchet the winch to tighten the hoist chain.
Before putting weight on the hoist chain, be sure that the links are aligned properly. This helps keep the chain from binding up, or becoming kinked, which is nearly impossible to correct when the chain is under tension.
14. Press the release tab and pull out the SRPS to release it from the rail.
15. Release the PEN cabinet outer slide rails and slide them back into the cabinet.
You may want to use a screwdriver to release the outer slide rail.

Illustration 4-10: Release Outer Slide Rail Locking Levers



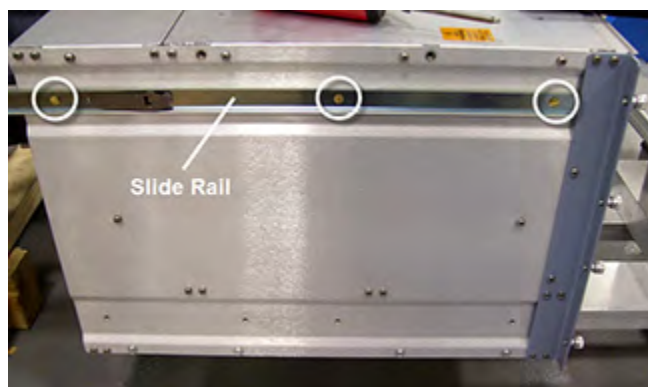
16. Lower the SRPS onto the FRU packaging platform for removal.

Illustration 4-11: SRPS FRU Packaging



17. Loosen the hoist chain and remove the spreader bar.
18. Remove the lifting brackets.
19. Remove both slide rails from the SRPS.

Illustration 4-12: Remove Slide Rails



20. Remove the winch by removing the pin from the main lifting beam and sliding out the winch. Place the winch (including hook and chains) into the hoist service kit box.

NOTICE

Before returning lifting beam to PGR cabinet, make sure that LOTO is still in place for PGR cabinet.

21. Remove the main lifting beam. Return the lifting beam to the PGR cabinet. Insert the hook end of the beam into the cabinet edge first, then place the bottom part of the beam in. Secure it into the cabinet with the Velcro straps and pin at the lower end of the beam.
22. After the main lifting beam has been inserted in the cabinet, plug in the XPS J9 connector on the Z-axis.
23. Unscrew the support tube from the top of the cabinet. Return the support tube to the HEC and secure with the two bolts.

2 Access Panel Upgrade



DANGER

STRONG MAGNETIC FIELD!

FERROUS MATERIALS CAN BECOME DANGEROUS PROJECTILES IN THE PRESENCE OF THE MAGNETIC FIELD PRODUCED BY THE MAGNET. DO NOT BRING ANY FERROMAGNETIC TOOLS OR EQUIPMENT INTO THE MAGNET ROOM.



DANGER

ELECTROCUTION HAZARD

HIGH VOLTAGE PRESENT

DANGEROUS VOLTAGES EXIST INSIDE THE PEN CABINET WHEN ENERGIZED.



WARNING

PERSONAL INJURY AND EQUIPMENT DAMAGE

STRONG MAGNETIC FIELD

WHEN SERVICING ANY MAGNETIC EQUIPMENT, IT IS CRITICALLY IMPORTANT THAT THE SERVICE ENGINEER CONSCIOUSLY PLAN THE PATH TO BE TAKEN WHEN MOVING HIGHLY FERROUS DEVICES IN THE MAGNET ENVIRONMENT. THE PATH SHOULD BE AS FAR FROM THE MAGNET AS PRACTICAL AND AVOID HIGH FLUX-DENSITY FIELDS.

SAFETY REQUIREMENTS:

- MOVEMENT OF FERROUS MATERIAL IN THE MAGNET ROOM MUST FOLLOW THE GE SERVICE PROCEDURE FOR THAT DEVICE. WHEN EXITING, MOVE AWAY FROM THE MAGNET IN THE MOST DIRECT MANNER POSSIBLE. EXCEPT WHEN MOVING FERROUS MATERIAL TO AND FROM ITS NATIVE LOCATION ON OR NEAR THE MAGNET, THE STATIC MAGNETIC FIELD IN ANY PORTION OF THE SERVICE PATH SHALL NOT EXCEED 200 GAUSS.
- TWO (2) MR SAFETY TRAINED PERSONNEL MUST BE PRESENT AT ALL TIMES WHEN SERVICING HIGHLY FERROUS DEVICES IN THE AREAS OF MAGNETIC FIELDS.
- WHEN PLANNING A SERVICE PATH, IT IS CRITICAL THAT THE PATH BE CLEAR AND SUFFICIENTLY WIDE. ENSURE THAT THERE ARE NO TRIP HAZARDS, OBSTACLES, CLUTTER, SLIPPERY SURFACES OR OTHER ITEMS EVEN PARTIALLY RESTRICTING THE PATH. IF THERE ARE PORTABLE OBSTACLES IN A PATH, REMOVE THEM FROM THE AREA AND REPLACE THEM AFTER THE SERVICE ACTION IS COMPLETED. IT IS REQUIRED TO WALK THE PATH PRIOR TO BEGINNING SERVICE TO ENSURE THAT THERE IS SUFFICIENT SPACE THROUGH WHICH TO PASS FOR YOU AND THE OBJECT BEING SERVICED.



WARNING

POSSIBLE PERSONAL INJURY!

THIS PROCEDURE REQUIRES MOVEMENT OF HIGHLY FERROUS MATERIAL IN CLOSE PROXIMITY TO THE MAGNET. ANY FIELD STRENGTH >200 GAUSS WILL FORCIBLY ATTRACT THE BLOWER BOX AND THE MOTOR PANEL ASSEMBLY.

STAY OUTSIDE THE 200 GAUSS LINE WHEN MOVING EITHER COMPONENT. KEEP AS FAR FROM MAGNET AS POSSIBLE AND REFER TO THE MR MAGNET SAFETY REQUIREMENTS IN THE MR SERVICE SAFETY MANUAL, PN 5452735.

2.1 Remove Access Panel

1. Disconnect all cables from magnet room side of SRPS Access Panel. As each cable is removed, confirm the J numbers on cable label matches the access panel labeling.

Illustration 4-13: SRPS Access Panel



2. Remove all fasteners attaching Access panel to PEN Panel.
3. Carefully remove Access Panel from PEN Panel. Cables on equipment room side are still attached to PEN cabinet modules.

NOTE: Access Panel cables attached to the rear or the SRPS module should have been disconnected already.

4. If site has MNS Amplifier, remove J33 from inside of the panel and set aside.
5. Disconnect the Access Panel cables from the following modules:

NOTE: Make sure that the sub-D filters connected to the PHPS2 module at J35, J36 and J54 are not removed. They need to stay in place and the new cables connected to them.

Table 4-2: Cable Disconnect Locations

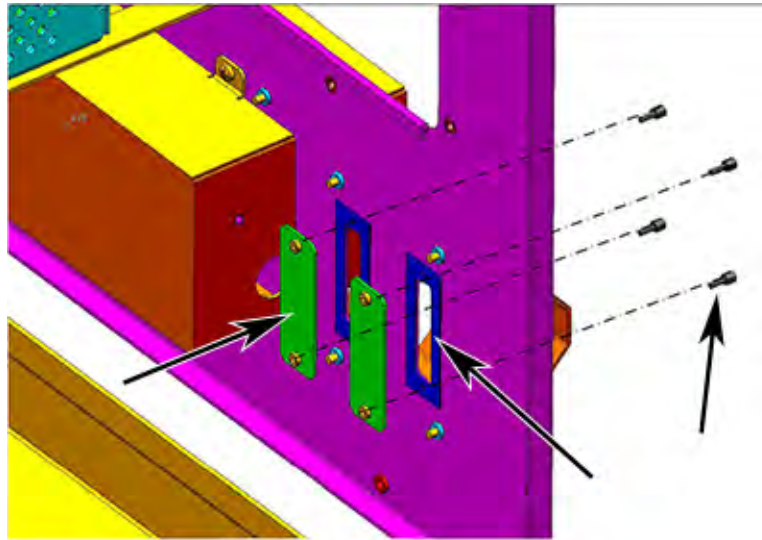
Module	Location
Exciter	J32
Exciter	J30
Exciter	J29
Exciter	J15
Driver	J17, J19
PHPS2	J35 (See Note)
PHPS2	J36 (See Note)
PHPS2	J39 (See Note)
PHPS2	J40 (See Note)
PHPS2	J41 (See Note)
PHPS2	J42 (See Note)
PHPS2	J54 (See Note)

NOTE: Remove and save the cylindrical ferrite clamp(s) from the cables. They **MUST** be installed on the new eSRPS cables.

The ferrite clamp is not a FRU. Take care when removing clamps so as not to damage the plastic clamp latch. If broken, use tie-wraps or some other method to install clamps on new cables.

6. Remove access panel from room and return for recycling.
7. Make sure cables at J51 and J52 on each side of PEN panel are disconnected and removed.
8. At J51 and J52, install EMI Gasket (5266899) on inside of panel along with Cover Plates (5442341) and Screwlocks (46-221417P2).

Illustration 4-14: J51 and J52 Cover Plates



2.2 Install eSRPS Access Panel

1. Carefully move new eSRPS Access Panel (5402530) into position near opening on PEN Panel on magnet room side.
2. If site has MNS amplifier, connect J33 cable to inside of panel.
3. Connect equipment room side cables to designated modules as indicated in table below. make sure that any ferrite clamps removed in previous procedure are reattached to the applicable cable connecting to the PHPS2 module. make sure that all sub-D filters are in place.

Table 4-3:

Module	Location	Sub-D Filter on PHPS2	Number of Ferrite Clamps
Exciter	J32		
Exciter	J30		
Exciter	J29		
Exciter	J15		
Driver	J17, J19		
PHPS2	J35 (See Note)	Yes	One
PHPS2	J36 (See Note)	Yes	One
PHPS2	J39 (See Note)		One
PHPS2	J40 (See Note)		Two
PHPS2	J41 (See Note)		One
PHPS2	J42 (See Note)		One
PHPS2	J54 (See Note)	Yes	One

NOTE: Attach the ferrite clamp(s) to the module end of the cables. Make sure that the same number of ferrite clamps are used on the new cable that were on the old cable.

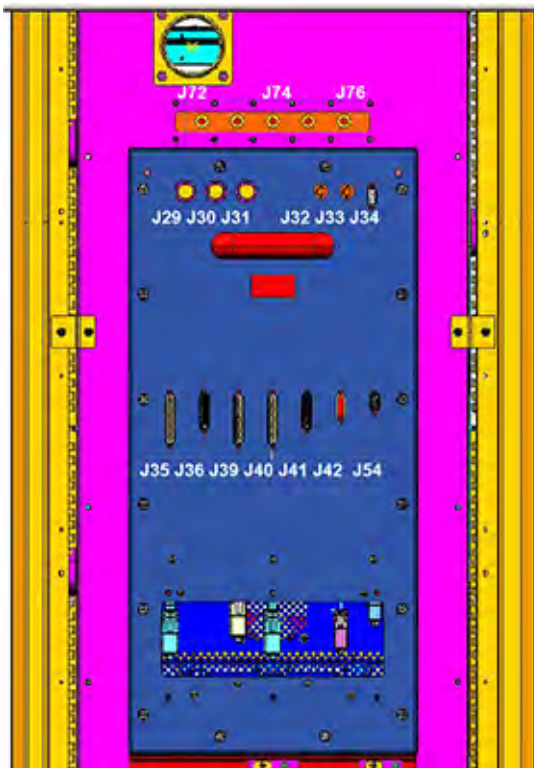
NOTICE

CAUTION: If ferrite clamp is broken, use tie-wraps or some other method to install clamps on new cables to make sure they remain permanently connected to cable and do not fall off.

Artifact and image degradation can result if clamps not applied correctly.

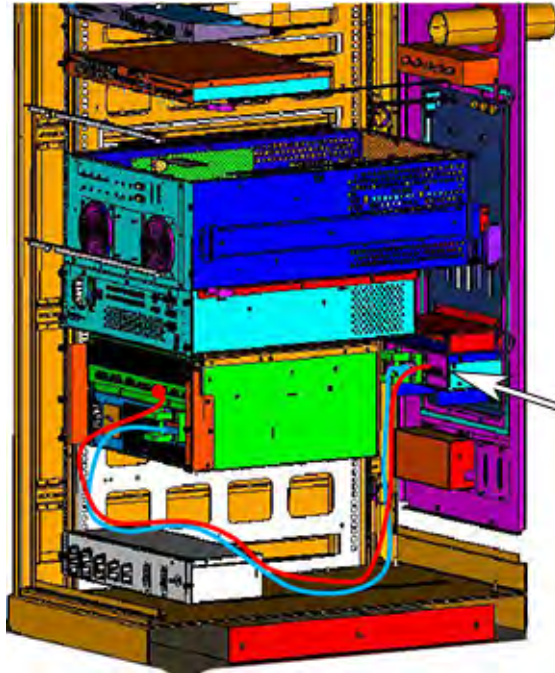
4. Attach new cables to connectors on inside of panel.
5. Attach the rear Access Panel to the PEN Panel.

Illustration 4-15: Access Panel Attachment



6. Route the two new cables (5575010 & 5575011), that will be attached to the front of the eSRPS, towards the Filter Box opening in the rear Access Panel.

Illustration 4-16: Filter Box Cables

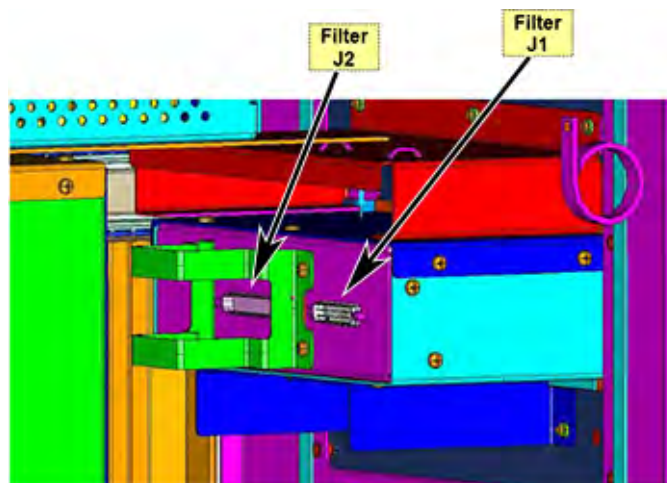


WARNING

POSSIBLE PERSONAL INJURY
THE FILTER BOX BEING INSTALLED IN THE FOLLOWING STEPS
INCLUDES SOME MAGNETIC PARTS.
WHEN MOVING THE FILTER BOX WITHIN THE MAGNET ROOM, KEEP IT
AS FAR AWAY FROM THE MAGNET AS POSSIBLE.

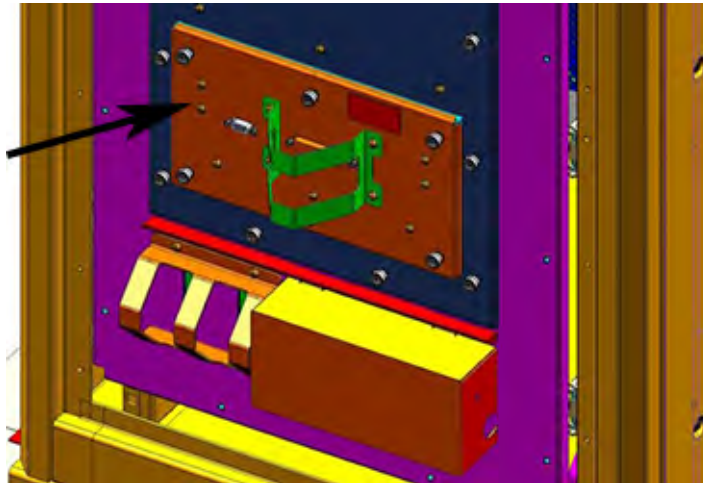
7. Move the eSRPS Filter Box (5402850) to opening and attach the two cables to J1 and J2 on the Filter Box.

Illustration 4-17: Filter Box Connections



8. Attach the Filter Box to the Access panel using the 6 captive screws.

Illustration 4-18: Attach Filter Box



9. Route Run M3375 from Rear Pedestal Hub, slot 12–J1, and connect to J112
10. Reconnect existing magnet room cables to new eSRPS Access Panel.

Illustration 4-19: Connect Cables to eSRPS Access Panels



3 Install eSRPS Module

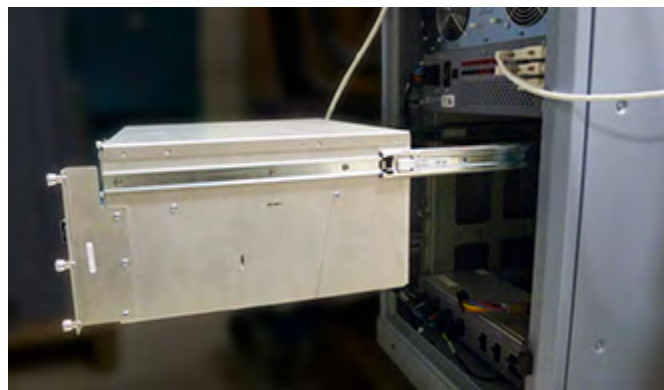
1. Attach both slide rails to the new eSRPS.

Illustration 4-20: Attach Slide Rails to SRPS Module



2. Connect the eSRPS rails to the mounting rails in the PEN cabinet, and slide the eSRPS into the cabinet.

Illustration 4-21: Install Module on Rails



3. Use the thumbscrews on the front of the eSRPS to secure the unit to the cabinet mounting rails.

Illustration 4-22: eSRPS Front



4. Connect the cables to the front of the eSRPS.

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Chapter 5 Magnet Room Upgrade Procedures

1 Magnet Enclosure Cover Removal

1.1 Overview

The magnet room upgrade consists of replacing the ODU cable, rear pedestal boards, and LPCA cable tracks. To perform this upgrade, enclosure covers are required to be removed.

1.2 Procedure

1. Make sure LOTO for the RF Amplifier and the PEN Cabinet has been performed. See Chapter 3, section 2.6 of the *MR Service Safety Manual*, PN 5452735.
2. Undock the patient table, and move it away from the magnet and out of the path between the dock and the magnet room exit door.

1.2.1 Remove Rear Pedestal Covers

1. Remove two screws on the side covers and one rear screw.
2. Lift up covers out of the bottom support brackets.

Illustration 5-1: Rear Pedestal Screw Locations



1.2.2 Remove Skirt Covers

Illustration 5-2: Screw Location - Front Cover

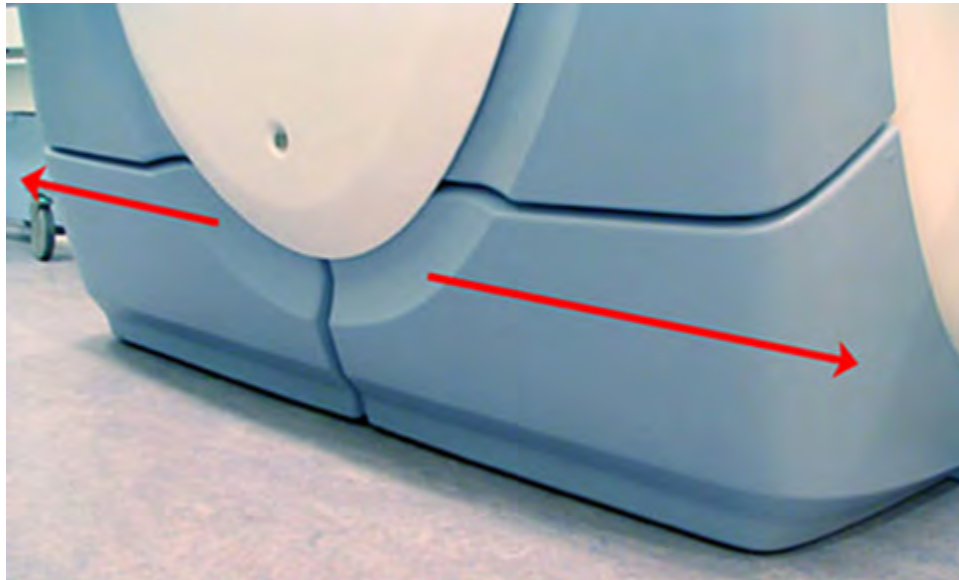


1. The skirt covers are held into place using poppers and screws. There is one screw that secures each skirt cover to the magnet. Remove the screw from the skirt cover being removed.

Illustration 5-3: Screw Location - Rear Cover



Illustration 5-4: Removing Skirt Covers



2. To remove a skirt cover simply slide it towards the front (or, in the case of the back covers, to the rear) of the magnet.

2 Dock ODU Cable Upgrade

2.1 Overview

This document contains procedures to remove the patient table dock, upgrade to the new ODU cable, and reinstall the dock.

The dock assembly is the most ferrous object in the magnet room. Utmost care must be taken when following these procedures. Never allow the end of the dock assembly (closest to the magnet) to come off the floor while in the magnet room. During the procedure, you will be asked to lift the front end of the dock assembly; the front end of the dock should never be lifted more than 3 inches (7.6 cm) while in the magnet room.



WARNING

POSSIBLE PERSONAL INJURY!

THIS PROCEDURE REQUIRES MOVEMENT OF FERROUS MATERIAL IN CLOSE PROXIMITY TO THE MAGNET.

TWO FIELD ENGINEERS ARE REQUIRED AT ALL TIMES.



WARNING

PERSONAL INJURY AND EQUIPMENT DAMAGE!

STRONG MAGNETIC FIELD.

WHEN SERVICING ANY MAGNETIC EQUIPMENT, IT IS CRITICALLY IMPORTANT THAT THE SERVICE ENGINEER CONSCIOUSLY PLAN THE PATH TO BE TAKEN WHEN MOVING HIGHLY FERROUS DEVICES IN THE MAGNET ENVIRONMENT. THE PATH SHOULD BE AS FAR FROM THE MAGNET AS PRACTICAL.

SAFETY REQUIREMENTS

- **THE SERVICE PATH MUST ALLOW FOR THE DOCK TO BE MOVED ALONG THE GROUND A DISTANCE EQUAL TO THAT OF THE DISTANCE BETWEEN THE MAGNET AND THE FOOT-END OF THE PATIENT TABLE PRIOR TO BEING LIFTED VERTICALLY.**
- **TWO (2) MR SAFETY TRAINED PERSONNEL MUST BE PRESENT AT ALL TIMES WHEN SERVICING HIGHLY FERROUS DEVICES IN THE AREAS OF MAGNETIC FIELDS.**

WHEN PLANNING A SERVICE PATH, IT IS CRITICAL THAT THE PATH BE CLEAR AND SUFFICIENTLY WIDE. ENSURE THAT THERE ARE NO TRIP HAZARDS, OBSTACLES, CLUTTER, SLIPPERY SURFACES OR OTHER ITEMS EVEN PARTIALLY RESTRICTING THE PATH. IF THERE ARE PORTABLE OBSTACLES IN A PATH, REMOVE THEM FROM THE AREA AND REPLACE THEM AFTER THE SERVICE ACTION IS COMPLETED. YOU MUST WALK THE PATH BEFORE BEGINNING SERVICE TO ENSURE THAT THERE IS SUFFICIENT SPACE THROUGH WHICH TO PASS FOR YOURSELF AND THE OBJECT BEING SERVICED.



WARNING

POSSIBLE PERSONAL INJURY!
STRONG MAGNETIC FIELD.

NEVER DISASSEMBLE THE DOCK IN THE MAGNETIC FIELD. REMOVE THE DOCK ASSEMBLY FROM THE MAGNET ROOM AS DESCRIBED IN THIS PROCEDURE BEFORE DISASSEMBLING THE DOCK.

2.2 Upgrade Procedure

2.2.1 Preparing Dock for Removal

1. Disconnect all cable connections from the magnet and the rear pedestal to the rear of the dock.
 - Systems that have the cable interface for receive cables on magnet front right foot
 - Systems that have the dock receive cables going all the way to the RF hub
 - Systems that have SRI cables going to magnet left front foot
2. Remove the associated cables that pertain to the site's system configuration. If site has I/O bracket on magnet front right foot, remove bracket and cables routed to rear pedestal.

Illustration 5-5: Right Side Bracket and Cables



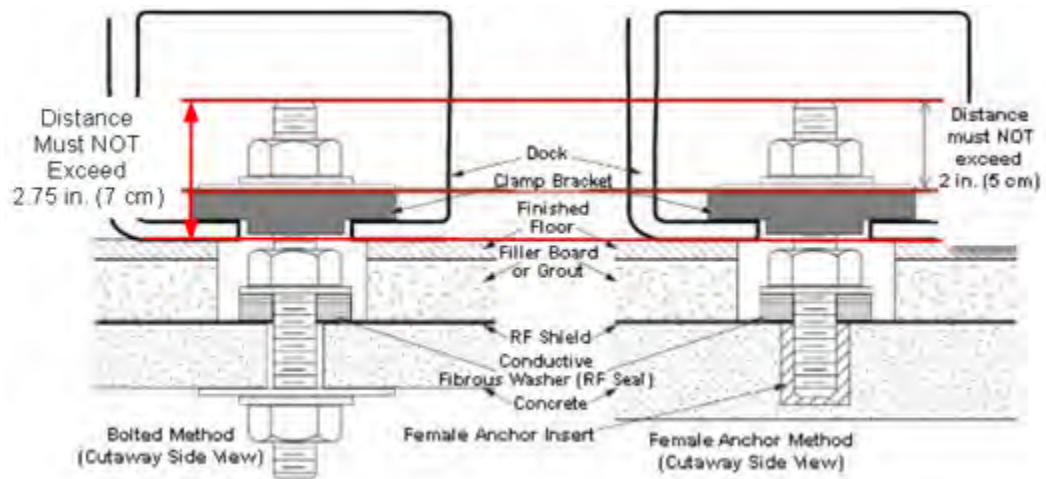
3. To assist in repositioning the dock during reinstallation, mark the placement of the dock on the floor. Depending on the site's need, place tape on floor at the leading edge of the dock, and on the right and left sides. Both are shown in the illustration.

Illustration 5-6: Tape to Mark Original Position of Dock



4. Visually check the height of the anchor stud that bolts the dock assembly to the floor. The height of the stud should not exceed 2.75 in (7 cm) as measured from the floor or 2 in (5 cm) as measured from the top of the clamp bracket.

Illustration 5-7: Measure Height of Anchor Stud



If you do not have something with which to measure the anchor stud, look straight into the opening on the right side of the dock assembly. If you cannot see the top of the stud, the stud is too tall and might need to be cut.

Illustration 5-8: Example of Stud That Must Be Cut



5. If the height of the stud does NOT exceed 2.75 in (7 cm) (as measured from the floor), proceed to [Section 2.2.3](#)

If the height exceeds 2.75 in (7 cm) (as measured from the floor):

- Loosen the stud anchor nut with a non-ferrous wrench and unscrew it a few turns. If the nut and washers are in the way, remove them.
- Try to remove the anchor stud with your fingers. Do not use any tools, because this could damage the anchoring system.
- If you cannot remove the anchor stud, try to lower the anchor stud by turning the stud to make it shorter. Use only your fingers to turn the screw, because a tool might damage the anchoring system.
- If you **are** able to remove or lower the stud, proceed to [Section 2.2.3](#).
- If you cannot remove the stud or lower the stud, order the Magnet Dock Bolt Cutting Kit area tool (PN 5435943) and proceed to [Section 2.2.2](#).

2.2.2 Cutting Anchor Stud



NOTICE

Make sure to wear gloves during the cutting procedure.

1. At the rear of the dock, use a $\frac{3}{4}$ inch non-ferrous wrench to LOOSEN (not remove) 4 nuts (2 on each side) from the dock bracket bolts.

Illustration 5-9: Dock Brackets



2. Loosen the nut on the stud anchor, if not already loosened.
3. Push the dock assembly as far to the left as possible, positioning the stud anchor close to the opening. The stud needs to be as accessible as possible to ease the cutting procedure.
4. Retighten the four (4) nuts on the bolts attaching the dock assembly to the magnet and, also, retighten the nut on the stud anchor. This prevents any movement of the dock assembly during the cutting procedure.
5. The stud removal kit has a selection of different nuts. Select a spare nut of the same dimension and threading as the anchor stud. Attach it to the stud on top of the original nut. The top of the second nut is used as the cutting mark.
6. If you do not have a spare nut that fits, remove the anchor stud nut, remove the washers, and thread the nut completely to the base of the stud. Then cut a small hole in the tack cloth and fit it over the stud. Place all the washers from the kit on the stud, over the tack cloth, as shown in the next illustration, to achieve the height of the additional nut.
7. Cut a small hole in the center of a tack cloth square and fit it over the top of the stud. The tack cloth will catch and retain the metal filings as the stud is cut.

Illustration 5-10: Second Nut on Anchor Stud



8. Insert a second tack cloth square into the entrance, covering as much of the area as you can. These pieces of tack cloth will help trap any metal filings.



WARNING

POSSIBLE PERSONAL INJURY!

THIS PROCEDURE REQUIRES MOVEMENT OF FERROUS MATERIAL IN CLOSE PROXIMITY TO THE MAGNET.

TWO FIELD ENGINEERS ARE REQUIRED AT ALL TIMES.



CAUTION

Possible Equipment Damage!

Cutting the stud creates metal filings.

As you cut the stud, avoid removing the blade to prevent metal filings escaping the hole. If you need to remove the blade, carefully use a clean piece of tack cloth to wipe the blade as you remove it from the hole.



CAUTION

Possible Equipment Damage!

Hacksaw blade is ferrous.

Perform all adjustments and blade replacements of hacksaw outside the magnet room. Make sure that the blade is secure in the hacksaw handle BEFORE entering the magnet room. Confirm that the wrist strap is secured on the wrist before entering the magnet room.

NOTE: If the blade on the hacksaw becomes dull during the cutting process, leave the magnet room, disassemble the hacksaw, and flip the blade so a new set of teeth are now cutting the stud.

9. Confirm that the end of the blade extends about 1.5 inches (3.8 cm) beyond the top edge of the hacksaw. This is the portion of the blade that you will use to cut the anchor stud.

Illustration 5-11: Extended Blade on Hacksaw



10. Cut the stud just above the top nut.
11. Carefully remove the pieces of tack cloth. Roll them up to contain any loose metal filings.
12. Use a clean piece of tack cloth to wipe the area, removing any metal filings that might be present.
13. Remove the cut end of the stud and the soiled tack cloth from the magnet room.
14. Remove the second nut.

If you used the washers from the kit, remove the washers and discard them. Place the original washer on the anchor stud and reattach the nut.

2.2.3 Removing Dock from Magnet Room

NOTE: Remove only the nuts from the rear of the dock that connect it to the dock brackets. You do not have to remove the dock brackets from the magnet.

1. At the rear of the dock, use a $\frac{3}{4}$ inch non-ferrous wrench to remove 4 nuts (2 on each side) from the dock bracket bolts. Make note of the number of washers between the dock mounting bracket and the dock at each stud location.

Illustration 5-12: Dock Bracket Nuts and Washers



2. Remove the nut, washers, and clamp bracket from the anchor stud.

If the site has followed the Pre-Installation Manual and the anchor is a two-part assembly, remove the anchor stud (bolt) from the female sleeve anchor, using hand-force to turn the stud. This allows the dock to remain flush with the floor during removal. **This is the preferred method.** If you can remove the anchor stud, proceed to .

Otherwise, the dock must be lifted slightly to clear the anchor stud, as described below.



! WARNING

BACK INJURY!

THE DOCK IS HEAVY, WEIGHING MORE THAN 50 POUNDS, AND IT CONTAINS FERROUS MATERIAL, MAKING IT STRONGLY ATTRACTED TO THE MAGNET.

TWO PEOPLE ARE NECESSARY TO MOVE THE DOCK OUT OF THE MAGNET ROOM TO PERFORM MAINTENANCE. FROM THIS POINT IN THE PROCEDURE, DOWNWARD PRESSURE MUST BE CONTINUOUSLY APPLIED TO THE DOCK TO PREVENT IT FROM BEING PULLED INTO THE MAGNET BORE.



! WARNING

POSSIBLE PERSONAL INJURY!

TO OPPOSE ANY PULL FROM THE MAGNET, FIELD ENGINEERS MUST APPLY CONTINUOUS DOWNWARD PRESSURE TO THE TOP OF THE DOCK, EVEN WHEN LIFTING THE DOCK IN THIS STEP.

DO NOT PLACE ANY PART OF THE HUMAN BODY IN THE PATH BETWEEN THE DOCK AND THE MAGNET.

3. Applying pressure to the top of the dock, completely remove all dock attachment bolts, then remove washers.

Make a note of the number of washers that are placed between the dock and the bracket, because they will need to be reinstalled in the same quantities and locations.

Illustration 5-13: Apply Pressure to Top of Dock and Remove Bolts



NOTE: Remove only the nuts from the rear of the dock that connect it to the dock brackets. Do NOT remove the dock brackets from the magnet.

4. Use two people to pull the dock away from the magnet and hold down the dock assembly.
 - a. Move the dock until the stud in the front is hitting the inner frame (labeled as 1)
 - b. Use two people to lift the front of the dock JUST ENOUGH to clear the top of the stud in the floor (labeled as 2)

Do not lift the front of the dock more than 3 inches (7.6 cm) from the floor. The rear of the dock must remain in contact with the floor.

- c. Use two people to move the dock to the left, just beyond the anchor stud (label as 3)
 - d. Use two people to lower the dock back on the floor immediately to the side of the anchor stud (labeled as 4)

Illustration 5-14: Pull Dock Away from Magnet, Lift to Clear Stud, Place on Floor to Side of Stud



5. With two field engineers, one holding each side of the dock, slowly and deliberately slide the dock away from the magnet, keeping downward pressure on the top of the dock assembly. Use care as the loose cables will be trailing behind the dock. Move the dock to a distance equivalent to where the foot of the patient table would normally be located.

Illustration 5-15: Apply Pressure to Top of Dock



6. When the distance between the dock and the magnet is at least equivalent to the distance between the magnet and where the foot of the patient table would normally be located, both field engineers may slowly lift the dock from the floor and exit the room.



CAUTION

Personal Injury!

The exposed stud poses a trip hazard.

Place a cone or similar item over the anchor stud (if not removable) to indicate a potential trip hazard.

7. Make sure to wear gloves during this cleaning procedure.

If the anchor stud was cut, return to the magnet room with a piece of tack cloth and clean the floor area where the dock was to remove any metal filings that may be present.

Also, clean the bottom of the dock assembly and the area of the anchor stud to remove any remaining metal filings before returning it to the magnet room.

2.2.4 Upgrading to New ODU Cable

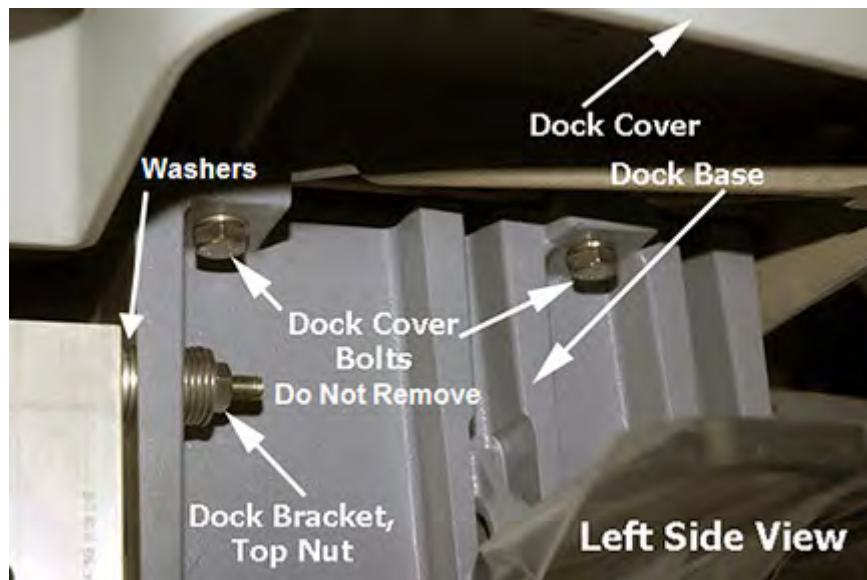


NOTICE

Be sure to disconnect the switch connector before removing the dock cover.

1. When the dock is outside of the Magnet Room, use a 3/4-inch wrench to remove four bolts (two on each side), and unfasten the dock cover from the dock base.

Illustration 5-16: Dock Cover Bolt Locations



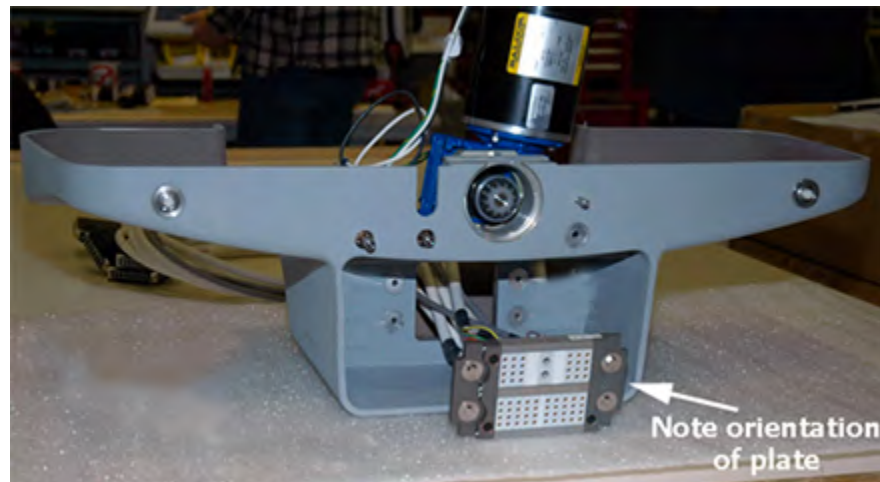
2. Turn the dock cover upside down as shown in the following illustration, being careful not to scratch the top of the cover, and set it on a protective cloth on the floor.
3. Using a 5/16-inch nut driver, remove the four nuts from the front of the Connector Plate.
4. Leave the Dock Connector Spacers attached to the dock.

Illustration 5-17: Remove Connector Plate



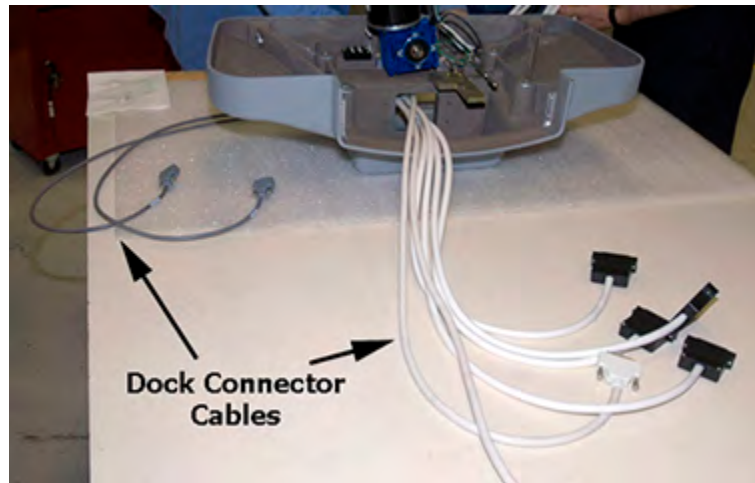
5. Carefully pull the Dock Harness cables out through the opening inside the dock cover and remove the dock Connector Plate assembly and cables.

Illustration 5-18: Removal of Cables



6. Locate the new ODU Dock harness (5443974). Route the four large black connectors (two at a time) through the opening in the dock cover, followed by the two large white connectors, and the remaining two gray cables.

Illustration 5-19: ODU Dock Harness Installation



7. Reattach the dock Connector Plate to the connector spacers on the front of the dock, paying attention to the correct orientation of the plate.

NOTE: Hold the dock spacer screws in place while tightening the dock Connector Plate to the connector spacers. To prevent damage, do not over tighten the M4 Dock Connector Bolts as they can easily shear.

Illustration 5-20: Attaching the ODU Connector Plate



8. Carefully turn the dock cover right side up, reconnect the dock switch connector, and place the dock back on the dock base. Reattach the dock bolts.

2.2.5 Reinstalling Dock



WARNING

POSSIBLE PERSONAL INJURY!

THIS PROCEDURE REQUIRES MOVEMENT OF FERROUS MATERIAL IN CLOSE PROXIMITY TO THE MAGNET.

TWO FIELD ENGINEERS ARE REQUIRED AT ALL TIMES.

NOTE: Any time you remove the dock from the magnet, you must check levelness and alignment of the dock and patient table when you reattach the dock to the magnet.

1. With two field engineers, one holding each side of the dock, enter the magnet room and place the dock on the floor where the foot of the patient table would normally be located.
2. With two field engineers, one applying vertical downward pressure to each side of the dock, slide the dock toward the magnet front.
3. As dock is being positioned, route ODU cables as indicated through Dock Mounting Bracket to cable bracket on left side magnet leg and to rear pedestal along right side of magnet.
4. Position the dock to clear the floor stud. There are two ways to do this:
 - a. If the site has followed the Pre-Installation Manual (PIM), then when reinstalling the dock it should be possible to remove the top portion of the stud. If this is the case, then continue to maintain pressure on the top of the dock and carefully slide the dock against the base of the magnet. Reinstall top portion of anchor stud.
 - b. If the site has an anchor stud that protrudes vertically from the floor and cannot be removed, a non-conformance MUST be raised with the customer and the anchor must be fixed to comply with the Pre-Installation Manual before any future servicing of the dock. To complete the task this time, follow these instructions per the reverse directions shown in the next illustration.
 - i. While maintaining pressure on the top of the dock, position the dock against the base of the magnet to the side of the floor stud at the front of the dock (number 3).
 - ii. Lift the dock just enough to clear the top of the stud in the floor (number 2).
 - iii. While rotating the dock, align the hole at the base of the dock with the stud and place the dock down into position and re-anchor the dock (number 1).

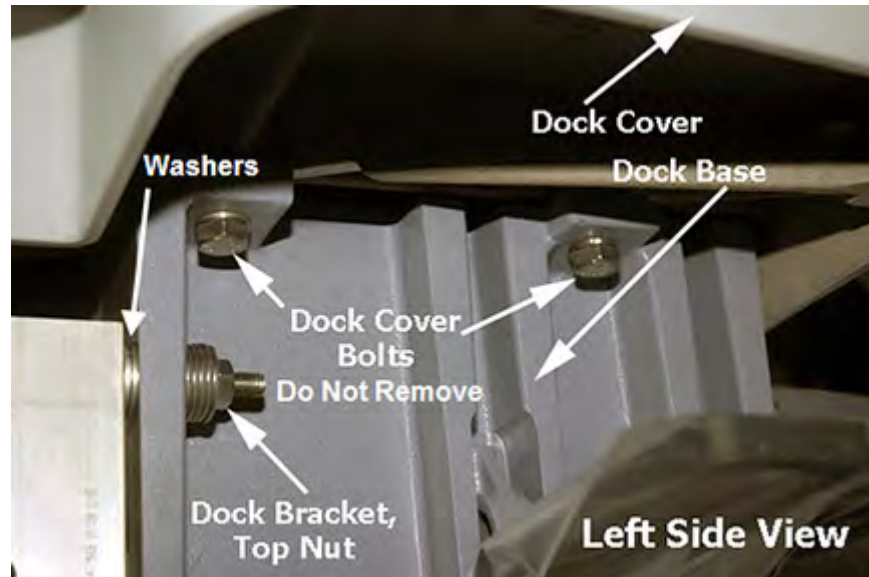
Illustration 5-21: Positioning Dock When Anchor Stud Cannot Be Removed



5. Reattach the nuts and washers to the dock bracket.

NOTE: Take care with the washers that are placed between the dock and the bracket. Ensure that the same number of washers is installed behind the dock on each bolt.

Illustration 5-22: Dock Brackets

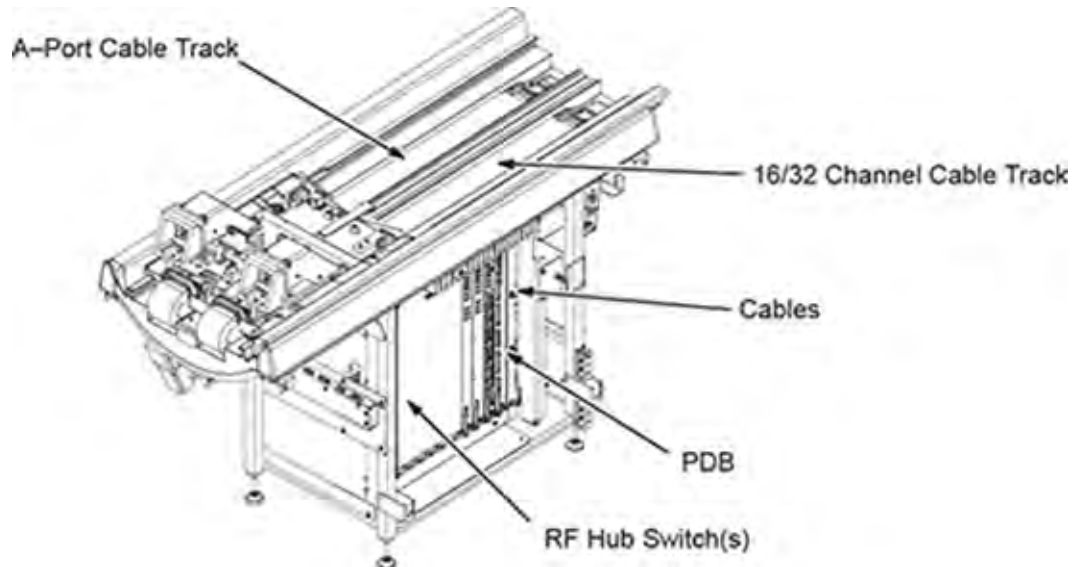


6. Connect all cables routed to rear pedestal.
7. After upgrade is complete and all cables have been routed, dressed, and connected — reinstall the magnet right and left side skirt covers.

3 Rear Pedestal Upgrade

The components in the following illustration of the rear pedestal will be upgraded to new components.

Illustration 5-23: Rear Pedestal Upgraded Components



3.1 RF Hub Boards



CAUTION

Possible Personal Injury

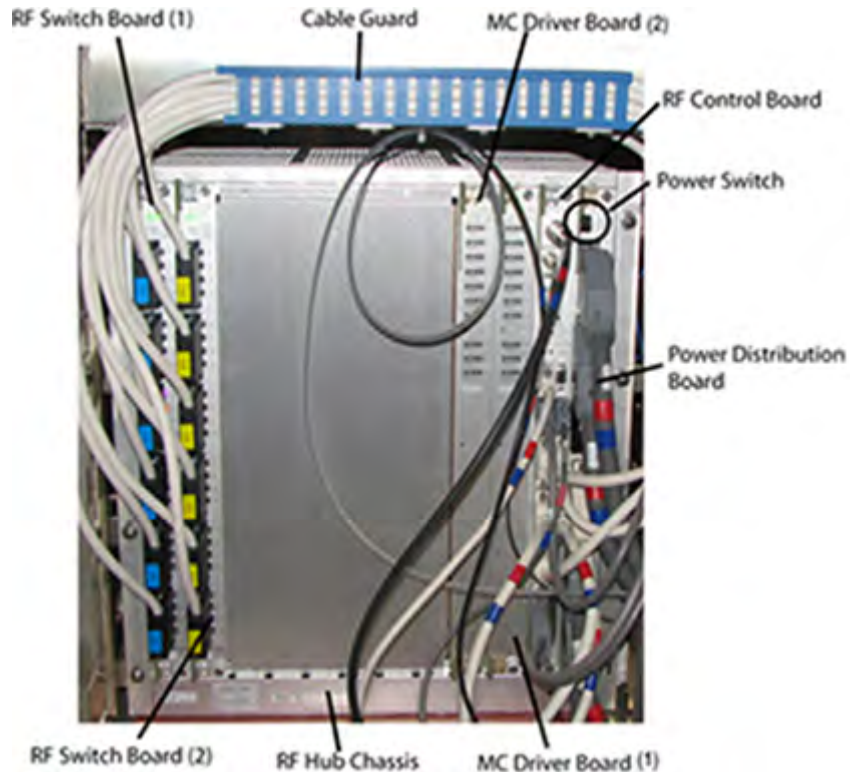
The boards being removed and installed in the following steps include some magnetic parts.

When moving the boards within the magnet room, keep them as far away from the magnet as possible.

3.1.1 Hub Board Removal

Make sure LOTO has been performed for the rear pedestal. See the MR Service Safety Manual, PN 5452735.

Illustration 5-24: Board Locations

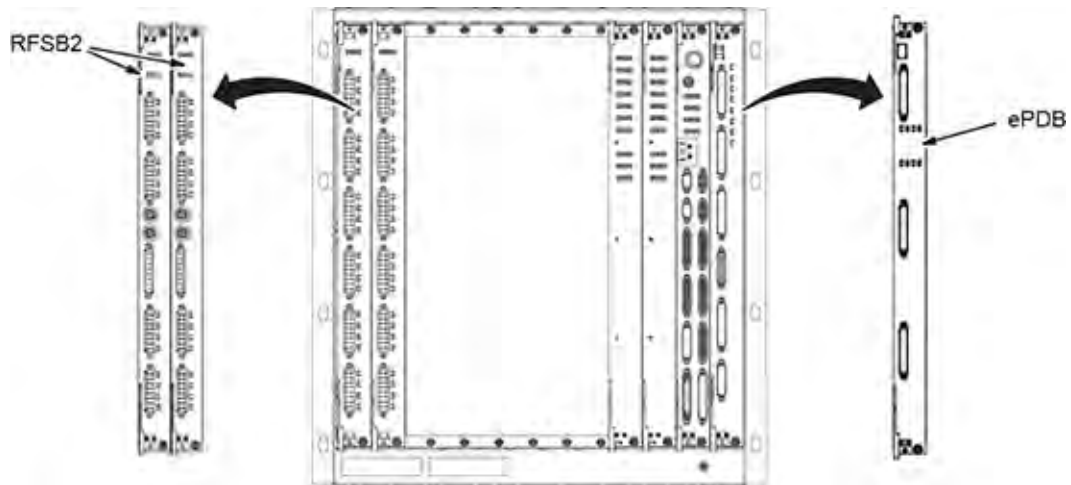


1. Disconnect cables routed from both cable tracks and also RFSB, slot 1 (and slot 2 if installed) and PDB, slot 12.
2. Loosen the top and bottom screws on front of the RFSB(s) and the PDB.
3. Lift up on the top and bottom release levers to eject the boards.
NOTE: The boards will not eject unless both screws are thoroughly loosened.
4. Slide out the boards from the RF chassis.
5. Place each of the boards in an ESD bag and remove them from the magnet room.

3.1.2 Installing New Hub Boards

1. Slide the two new RFSB2 (6300662-2) into slots 1 and 2. Make sure the boards are in the track.
2. Slide the new ePDB (6850022-2) into slot 12. Make sure the board is in the track.

Illustration 5-25: New Boards



3. Line up the mechanical lever groove with the chassis edge, and lower the levers to insert the board.

Illustration 5-26: Groove on RF Board Lever



NOTE: Make sure the board is fully seated in the RF chassis.

4. Tighten the top and bottom screws to secure the boards to the RF chassis.
5. Attach cables to all boards when upgrade of rear pedestal is complete.

3.2 Cable Tracks



WARNING

FERROUS MATERIAL HAZARD!

THE CABLE CONNECTORS ON BOTH TRACKS CONTAIN FERROUS MATERIAL.

FOLLOW ALL GE MAGNET SAFETY GUIDELINES WHEN WORKING CLOSE TO THE MAGNET BORE. SEE THE MR SERVICE SAFETY MANUAL, PN 5452735.

3.2.1 Removal of P1/P2 Cable Track

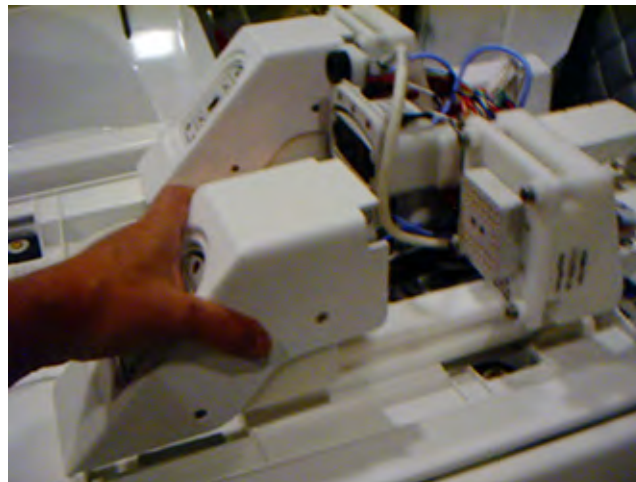
1. Remove LPCA cover by unfastening the four screws.

Illustration 5-27: LPCA Cover Screws



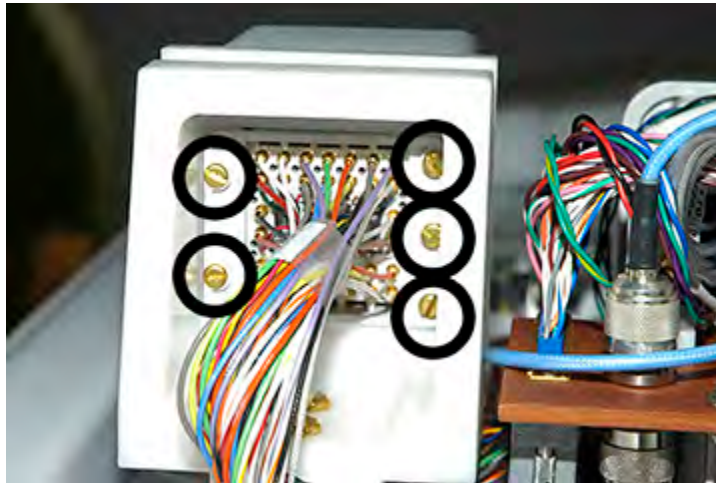
2. Remove the P1 and/or P2 quick release modules. Use sufficient force to pull the module forward to slide it off the track.

Illustration 5-28: Remove the Quick Release Module



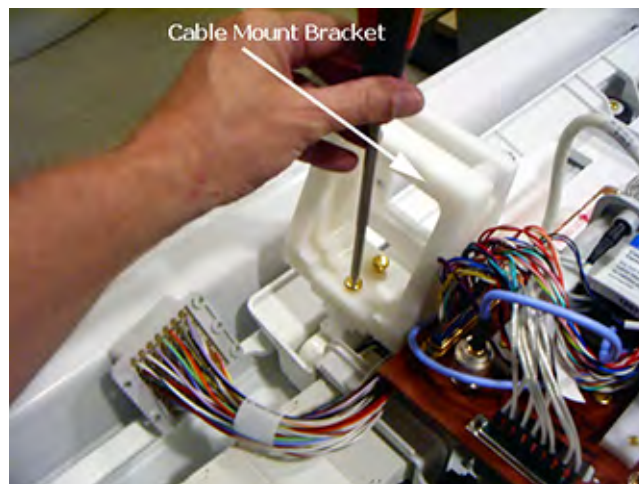
3. To create enough space to remove and replace P1/P2 cable connectors, do the following:
 - a. At the rear of each cable mount bracket, remove two screws from the left side and three screws from the right side, and then set the cable connector aside.

Illustration 5-29: Cable Connector Screws



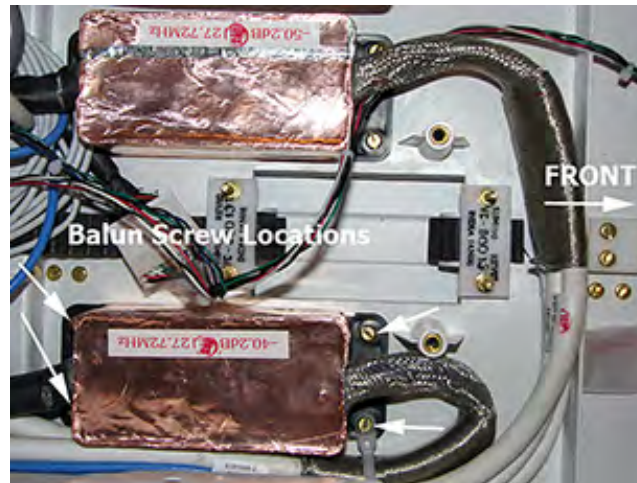
- b. Remove both bottom cable mount bracket screws, and then set the cable mount bracket aside.

Illustration 5-30: Cable Mount Bracket Screws



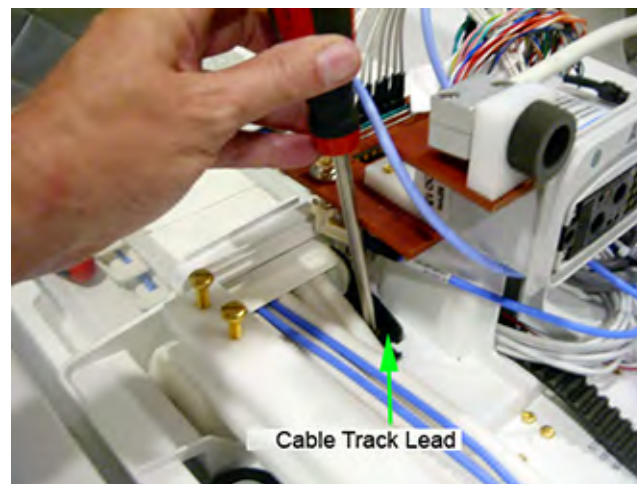
4. Remove four slotted brass screws from the front and back of the both baluns.

Illustration 5-31: Balun Screw Locations



5. Remove the two slotted head brass screws that secure the cable track lead to the base of the LPCA.

Illustration 5-32: Cable Track Lead Screw Locations



6. Remove the two slotted head screws that secure the cable track lead to the front of the rear pedestal bridge.

Illustration 5-33: Bridge Cable Track Screw Locations



7. Disconnect all cable connections that feed from the front end of the cable track to points throughout the rear pedestal.

NOTE: Take note of cable routing and tie wrap locations.

Table 5-1: 16-Channel or 32-Channel LPCA Cable Track Connection Points

16-Channel or 32-Channel Cable Track Assembly	Other End Positions
PED-LPCA J5	PED J1 blue cable (on the right side as viewed from the patient table)
	PED load B blue cable (at the back of the rear pedestal)
	P1 A (reroute box), or PED-Hub slot 1 P1
	P1 B (reroute box) 32-channel only, or PED-Hub-slot 2 P1
	PED-Hub-slot_11 J1, on RFCB slot 11 in RF hub
PED-LPCA J4	P2 A (reroute box), or PED-Hub-slot 1 P2
	P2 B (reroute box) 32-channel only, or PED-Hub-slot 2 P2
	PED-Hub-slot_11 J2, or RFCB slot 11 in RF hub

Table 5-2: 32 Channel P-Port Cable Track Connection Points (No Reroute Box)

LPCA End	Pedestal End
PED-LPCA J5	PED J1, blue cable (on the right side)
	PED Load B, blue cable (at the back of the rear pedestal)
	PED-Hub-slot1-P1
	PED-Hub-slot2-P1
	PED-Hub-slot 11 J1, on RFCB slot 11 in RF hub

8. Feed loose end of cable connections up through the opening at the front of the rear pedestal bridge and let them lay freely on the bridge.

Illustration 5-34: Cable Track Opening in Bridge



9. Starting at the rear of the LPCA, pull cable connectors through the cable well of the bridge, under the LPCA, and out of the back of rear pedestal. Allow the cable track to hang freely.
10. Starting from the front of the LPCA, pull the cable track (the side that includes the baluns) through the opening on the LPCA in a forward direction.

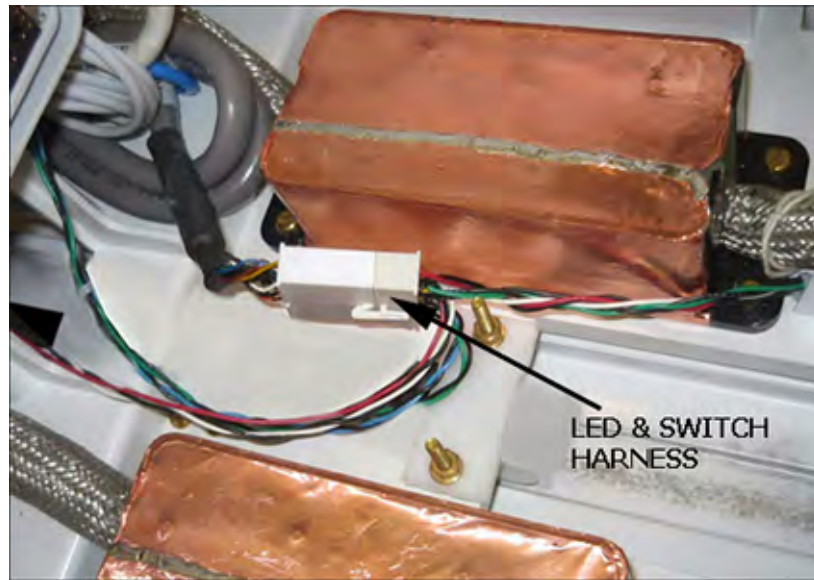
Illustration 5-35: Removal Path of P1/P2 Cable Track



3.2.2 Removal of A-Port Cable Track

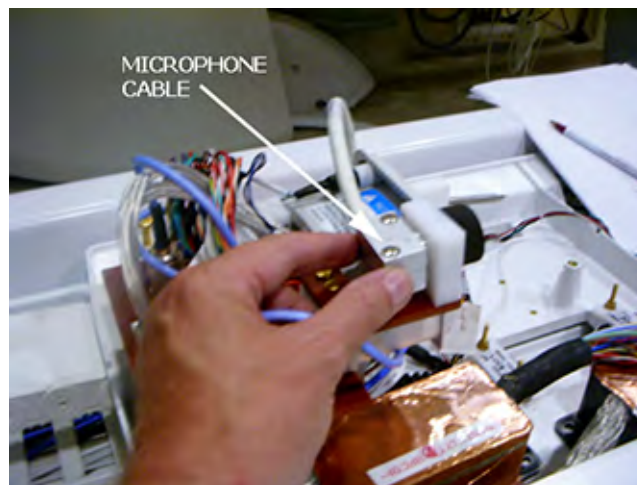
1. To create enough space to remove and replace the A-port connector cable track, remove the two slotted brass screws from the bottom of the P2 cable mount bracket as shown in [Illustration 5-30](#) and set it aside.
2. Disconnect the LED and switch harness connector.

Illustration 5-36: LED and Switch Harness



3. Remove the microphone cable from the rear of the microphone.

Illustration 5-37: Microphone Cable



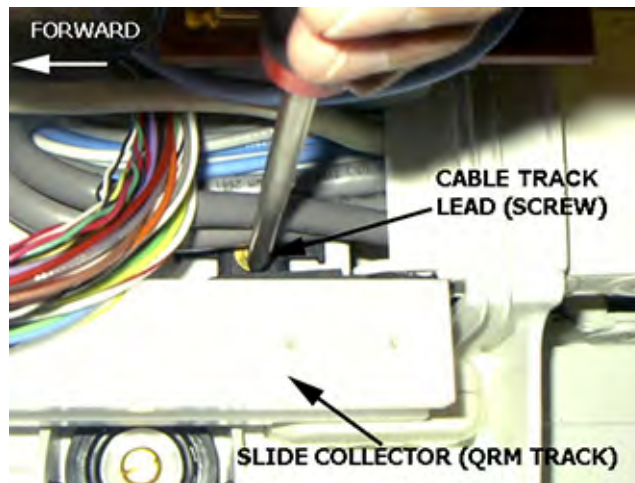
4. Remove four slotted brass screws from top of LPCA bulkhead plate. Carefully move wires out of the way when removing screws.

Illustration 5-38: Top of Bulkhead Plate



5. On the bottom of the bulkhead plate, remove the J1 connector first. Tilt the plate up to allow easier access to the retaining screws of J1.
6. With the plate still tilted, remove the remaining connectors (J2, J4, and J5) from the bottom of the plate.
7. Remove the two slotted head brass screws that secure the cable track lead to the base of the LPCA.

Illustration 5-39: Cable Track Lead Screws



8. At the back of the LPCA, pull the cable track through the cable frame (opening) of the LPCA assembly and let it hang freely.
9. Remove the two standard head screws that secure the cable track lead to the front of the rear pedestal bridge. See [Illustration 5-34](#)

10. Disconnect all cable connections that feed from the front end of the cable track to points throughout the rear pedestal.

Table 5-3: Cable Track Assembly Function and Other End Connection Points

LPCA Connection Location	Other End Position
PED-LPCA J1	Reroute box J8, OR PED-hub slot 1-A
PED-LPCA J5	PED J2 blue cable (on the right side as viewed from the patient table)
PED-LPCA J3	PED J1 gray cable (J1 on the table interface box, LPCA interface)
PED-LPCA J2	PED-hub slot 11 J6 (on RFCB in RF hub)
Microphone	J31 SRI (left side of magnet)
PED-LPCA J4	Dummy load A (at back of rear pedestal, opposite side of table interface module)

Table 5-4: 32 Channel A-Port Legacy Cable Track Connection Points

LPCA End	Other End
PED-LPCA J1	PED-Hub-slot1 A
PED-LPCA J5	PED J2, blue cable (on the right side)
PED-LPCA J2	PED-Hub-slot_11 J6
Rear microphone	Run M3346 MAG-SRI-J31
PED-LPCA J4	PED-Load-A, at the back of the rear pedestal
PED-LPCA J3	PED-TIM J1 (LPCA LED and switch cable)

11. Feed the loose end of cable connectors up through the opening at the front of the rear pedestal bridge and let it lay freely on bridge.

NOTE: Maneuver the J6 connector so that it is level (straight) and angled in a diagonal to pass through the opening in the bridge.

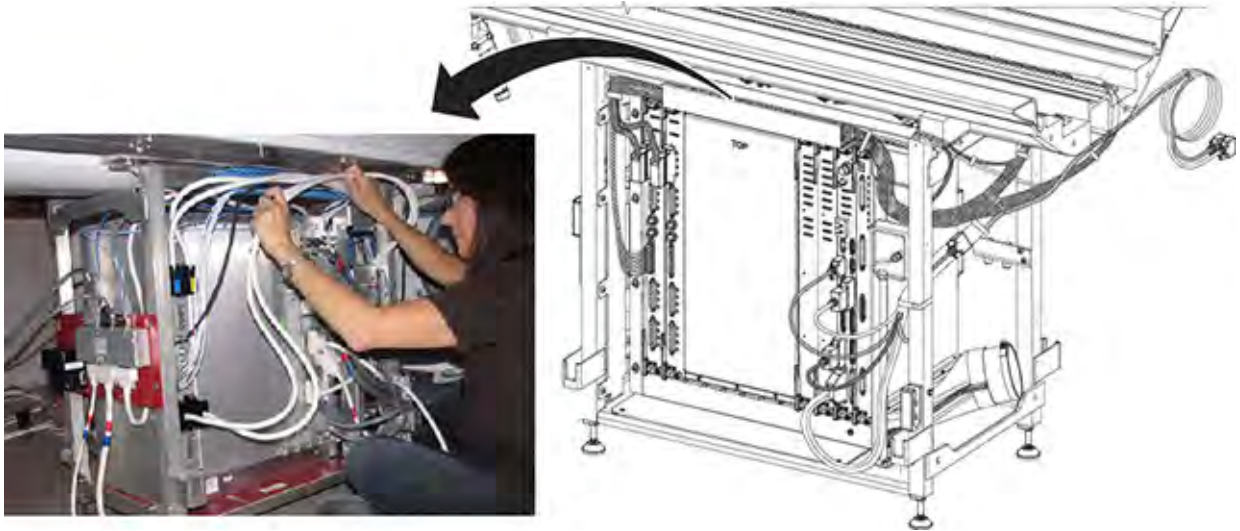
12. At the front of the rear pedestal bridge, pull the cable track through the cable well until the cable clears the underside of the LPCA.

3.2.3 Installation of new Cable Tracks

1. Move the replacement cable tracks into position.
2. To ensure proper installation and reconnection of cables, reverse the *Removal* procedures above starting with the A-Port cable track and then the P1/P2 cable track.

NOTE: Wire tie the cables out of the way of the quick release module and slide mounting. The quick release module must slide freely without interference.

Illustration 5-40: LPCA Cable Routing and Connections



3. Slide the quick release module to ensure it is fully inserted and there is no interference from the cable installation.
4. Fasten the four screws that secure the track assembly to the bridge and LPCA.
5. Reinstall the LPCA cover.

3.3 Dock Cable Routing and Connections

Route the cables from the new Dock ODU cables along the right side of magnet and connect to rear pedestal as marked.

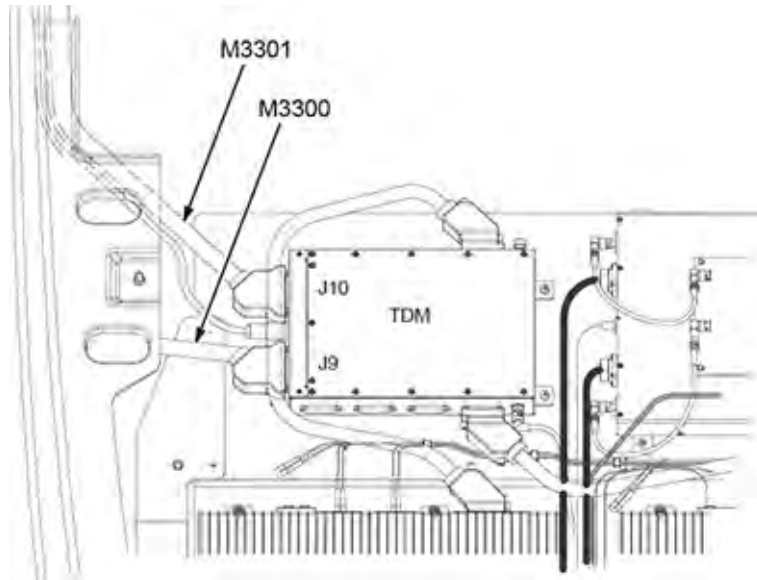
Illustration 5-41: ODU Cables at Rear of Magnet



3.4 TDM Cable Upgrade

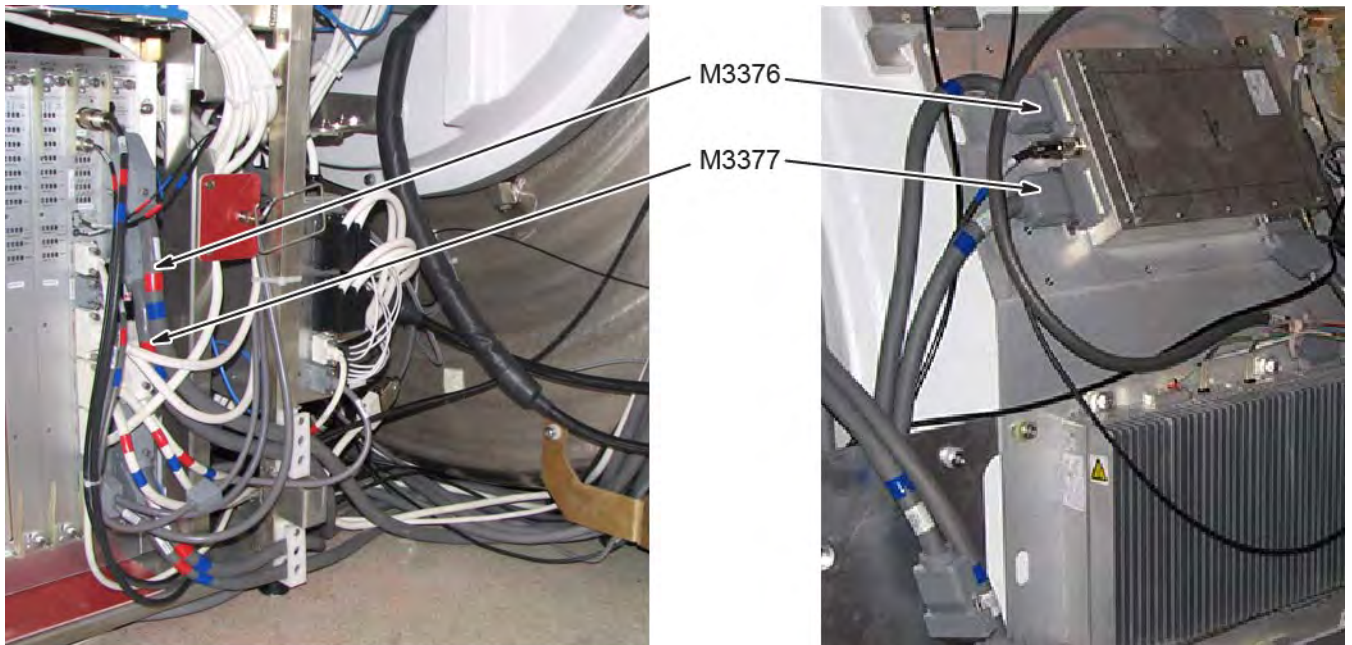
1. Disconnect Runs M3300 and M3301 from the TDM module.

Illustration 5-42: Disconnect Cables



2. Disconnect other ends of Runs M3300 and M3301 from J51 and J52 on the PEN Panel. Remove and discard cables.
3. Route and connect new cable Runs M3376 and M3377 from TDM J9 and J10 to Rear Pedestal ePDB slot J12, J2 and J3.

Illustration 5-43: Install New Cables



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Chapter 6 Power On Procedures and Finalization Steps

1 Removing LOTO

Remove LOTO from the system when upgrade is complete. See the *MR Service Safety Manual*, PN 5452735.

2 Finalization

1. Power on host.
2. Perform MCR-V checks from service procedures.
3. Perform Check Scan from service procedures.
4. Perform SaveINFO from service procedures.
5. Hand over.

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